

# Enalapril mitigates aging by targeting antioxidative genes via phosphorylated Smad1/5/9

Reviewed Preprint

v1 • February 12, 2025

Not revised

Wencong Lyu, Haochen Wang, Zhehao Du, Ran Wei, Jianuo He, Fanju Meng, Jinlong Bi, Lijun Zhang, Chao Zhang, Yiting Guan, Wei Tao ✉

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China • Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China • Zhanjiang Institute of Clinical Medicine, Central People's Hospital of Zhanjiang, Guangdong Medical University, Zhanjiang, China

 [https://en.wikipedia.org/wiki/Open\\_access](https://en.wikipedia.org/wiki/Open_access)

 Copyright information

## eLife Assessment

The manuscript presents **valuable** data suggesting that enalapril elevates pSmad1/5/9 signaling, reduces cellular senescence, and enhances physiological functions in aged mice. However, the in vivo evidence remains **incomplete**, as studies blocking pSmad1/5/9 or using NAC to negate enalapril's purported benefits are lacking, and no lifespan extension data are shown. While in vitro findings support Smad1/5/9 as a key mediator, additional experiments - including BMP receptor inhibition and comprehensive senescence markers - are necessary to validate its essential role in vivo. Overall, the study provides promising insights into enalapril's anti-senescence potential but requires further rigorous investigation to fully substantiate its mechanism and therapeutic impact.

<https://doi.org/10.7554/eLife.104774.1.sa3>

## Abstract

Aging increases the risk of a myriad of chronic diseases, which are expensive and difficult to treat owing to their various risk factors. Repurposing existing anti-aging medications has accelerated the development of therapies aimed at slowing aging. In this study, we revealed that enalapril, a drug widely prescribed for hypertension, can improve both cellular senescence and individual health. Mechanistically, phosphorylated Smad1/5/9 act as pivotal mediators of the anti-aging properties of enalapril. It stimulates downstream genes involved in cell cycle regulation and antioxidative defense, facilitating cell proliferation and diminishing the production of reactive oxygen species (ROS), thus increasing the antioxidative ability of enalapril. At the organismal level, enalapril has been shown to bolster the physiological performance of various organs; it notably enhances memory capacity and renal function and relieves lipid accumulation. Our work highlights the potential of enalapril to augment antioxidant defenses and combat the effects of aging, thereby indicating its promise as a treatment strategy for aging-associated diseases and its use for healthy aging.

## Introduction

Aging is a multifaceted process influenced by numerous factors, and its intricate mechanisms are not yet fully understood. Geroscience is a burgeoning field of study in which researchers hypothesize that aging is a significant risk factor for many chronic diseases. These findings suggest that interventions targeting the mechanisms of aging might delay the onset or slow the progression of these diseases(Kirkland, 2016 [↗](#); Lopez-Otin, Blasco, Partridge, Serrano, & Kroemer, 2023 [↗](#)). Cellular senescence, a pivotal indicator of aging, is characterized by irreversible cell cycle arrest triggered by cellular damage or stress(Khosla, Farr, Tchkonina, & Kirkland, 2020 [↗](#)). During cellular senescence, cells exhibit increased activity of the lysosomal enzyme senescence-associated  $\beta$ -galactosidase (SA- $\beta$ -gal), decreased proliferation marker Ki67 signals and increased senescence-associated secretory phenotype (SASP) secretion(Coppe et al., 2008 [↗](#); Dimri et al., 1995 [↗](#)).

Senescence is the result of a complex interplay of endogenous and exogenous signaling pathways(Cohen et al., 2022 [↗](#)). Currently, the activation of tumor suppressor pathways, such as p16<sup>INK4a</sup>/RB, is deemed essential for senescence-related cessation of proliferation(Baker et al., 2011 [↗](#)). The disruption of signaling pathways disrupts tissue homeostasis, leading to proinflammatory responses that are increasingly recognized as fundamental aspects of senescence(Coppe et al., 2008 [↗](#); Shah et al., 2013 [↗](#)). Researchers have suggested that the expression of p16 is suppressed by inhibitor of DNA-binding (ID) proteins. The expression of the *ID1* gene is downregulated by transforming growth factor- $\beta$  (TGF- $\beta$ ), while it is upregulated by BMP-mediated phosphorylated SMAD signaling in various cell types(Hayashi et al., 2016 [↗](#); Liang, Brunicardi, & Lin, 2009 [↗](#)). SMADs are crucial transcriptional effector factors responsible for transmitting TGF- $\beta$ /BMP signals(Urist, 1965 [↗](#)). They convey signals from the TGF- $\beta$  superfamily and homomeric serine/threonine kinases on the cell surface to the nucleus, directly regulating transcriptional programs critical for key events(Miyazono, Maeda, & Imamura, 2005 [↗](#)). Numerous studies have shown that the BMP-SMAD signaling pathway can modulate genes involved in the cell cycle, which affects the balance between cell proliferation and senescence(Genander et al., 2014 [↗](#); Hayashi et al., 2016 [↗](#)). Phosphorylated SMAD plays a crucial role in the induction and maintenance of pluripotency, helping maintain tissue homeostasis during aging(Orlowski, 2017 [↗](#)). However, dysregulation of BMP signaling during aging can lead to fibrosis, impaired wound healing, or excessive tissue calcification, which are detrimental to tissue function(Lyu et al., 2018 [↗](#); Rahman, Akhtar, Jamil, Banik, & Asaduzzaman, 2015 [↗](#)).

Owing to increasing insights into cellular and individual aging, novel aging mechanisms and signaling pathways have been identified, leading to the development of new therapeutic interventions. Senescent cells have recently been targeted in therapy given their substantial impact on aging (Childs et al., 2017 [↗](#)). Consequently, eliminating senescent cells through pharmacological interventions, known as “senolytics”, may delay the onset or slow the progression of age-related diseases and enhance physiological function(Childs et al., 2017 [↗](#)). Research has proven that specific senolytic drugs, such as dasatinib and quercetin, can eliminate senescent cells, reverse aging phenotypes and increase the survival rate of mice(Xu et al., 2018 [↗](#); Zhu et al., 2015 [↗](#)). However, the current senolytics are not universally effective across different cell types and display inconsistent sensitivities, thereby limiting their broad application(Yousefzadeh et al., 2018 [↗](#); Zhu et al., 2016 [↗](#)). In addition to senolytics, researchers have explored pharmacological interventions that aim to ameliorate or reverse aging phenotypes without the need to eliminate senescent cells. This included both the repurposing of existing medications and the development of new ones(Bharath et al., 2020 [↗](#); Harrison et al., 2014 [↗](#)). However, repurposing is particularly attractive due to its cost-effectiveness; these drugs have already undergone rigorous safety evaluations, allowing for quicker implementation in new contexts. For example, drugs such as metformin, which is typically used to treat type 2 diabetes,

and rapamycin, an immunosuppressant, have shown promise in reducing aging phenotypes and extending the lifespan (Bharath et al., 2020 [↗](#); Harrison et al., 2009 [↗](#)). This strategy is an innovative pharmacological intervention in which existing compounds target the aging process.

In this study, we employed a drug repurposing strategy to evaluate the anti-aging potential of several FDA-approved drugs and identified enalapril as a promising candidate. Our investigation revealed that enalapril increases pSmad1/5/9 levels, which in turn upregulates key downstream genes involved in cell cycle regulation and antioxidative defense, thereby improving cell proliferation, reducing inflammation, and decreasing reactive oxygen species (ROS) levels. Our findings confirmed the anti-aging mechanism of enalapril at both the cellular and organismal levels. Enalapril has the ability to delay the aging of various organs, with notable effects on the brain, kidney, and liver. Furthermore, treating aged mice with enalapril improved spatial memory and anxiety behavior, alleviated lipid accumulation, and slowed kidney aging. Taken together, the potential of enalapril to extend the lifespan and ameliorate age-related diseases makes it a compelling candidate for research and development in the field of anti-aging therapeutics.

## Materials and methods

### Cell culture

IMR90 cells were cultured in DMEM (Gibco, C11995500) supplemented with 10% fetal bovine serum (Cellmax, SA111) and 1% non-essential amino acids (Gibco, 11140050) at 37°C and 5% CO<sub>2</sub>. When the cell density in the in vitro culture reached 80–90% (usually within 2 or 3 days), cell passage was performed.

### Drug preparation and addition

Enalapril (MCE, HY-B0331) was dissolved in DMSO to prepare a 10 mM stock solution. When used, it was diluted to 10 μM in the culture medium. After the cells adhered to the surface, enalapril was added for treatment, and RNA and protein were collected for analysis. Recombinant BMP4 protein (MCE, HY-P7007) was dissolved in sterile ddH<sub>2</sub>O to prepare a 100 μg/mL stock solution. When used, it was diluted to 50 ng/mL in the culture medium and added to the cells. The BMP receptor inhibitor (LDN193189, MCE, HY-12071) was dissolved in sterile ddH<sub>2</sub>O to prepare a 1 mM stock solution. When used, it was diluted to 1 μM or 0.5 μM in the culture medium. The ID inhibitor (Pimozide, MCE, HY-12987) was dissolved in DMSO to prepare a 10 mM stock solution. When used, it was diluted to 5 μM in the culture medium. After the cells adhered to the surface, drugs were added for a 3-day treatment, and samples were collected for analysis.

### Senescence-associated β-galactosidase (SA-β-gal) Staining

SA-β-gal staining was performed using a Senescence β-Galactosidase Staining Kit (Sigma-Aldrich, CAS0030). Briefly,  $0.6 \times 10^5$  cells were seeded in a 6-well plate. The cells were fixed with fixation buffer and incubated at room temperature for 10 minutes. After the cells were washed three times with PBS, the prepared staining mixture was added to the 6-well plate, and the plate was incubated at 37°C overnight. The stained cells were observed and photographed under a microscope.

### Western blot

Proteins were extracted using TRIzol reagent (Invitrogen, 15596018CN) and dissolved in 1% SDS, after which SDS-PAGE was performed. The samples were transferred to nitrocellulose membranes and incubated with 5% skimmed milk at room temperature for 30 minutes. The primary

antibodies were incubated overnight at 4°C, followed by three washes with PBST. The membranes were then incubated with secondary antibody at room temperature for two hours. The band signals were visualized using an Odyssey Infrared Imaging System (Odyssey, LI-COR).

## RT-qPCR

Total RNA from cultured cells was extracted using TRIzol reagent (Invitrogen, 15596018CN) according to the manufacturer's instructions. The DEPC-treated water-dissolved RNA was subsequently reverse transcribed into cDNA using the HiScript III All-in-one RT SuperMix Perfect for qPCR (Vazyme, R333) following the product manual. The cDNA levels of specific primers were analyzed using the LightCycler<sup>®</sup> 96 qPCR system (Roche), and the results were normalized to those of  $\beta$ -actin. The primers used in this analysis are listed in Table S1.

## RNA-seq and data processing

Total RNA was extracted using TRIzol reagent (Invitrogen, 15596018CN) following the manufacturer's instructions. The extracted total RNA was then sent to Novogene for subsequent cDNA library construction. The library was sequenced with paired-end reads on the Illumina HiSeq PE150 platform.

For data processing, FastQC (version 0.11.5) software was first used for data quality control. Raw paired-end sequencing reads were trimmed by Trim Galore (version 0.6.10) software to remove adapter sequences and low-quality reads. The high-quality RNA-seq reads were mapped to the hg38/mm10 genome using HISAT2 (version 2.2.1) software(Kim, Paggi, Park, Bennett, & Salzberg, 2019). SAMtools (version 1.17) software was then used to convert the generated SAM files to bam files(Li et al., 2009). To qualify gene expression, the number of reads mapped to each gene was counted using featureCounts (version 2.0.1) software(Liao, Smyth, & Shi, 2014). Differentially expressed genes (DEGs) were calculated by R package DESeq2 (version 1.34.0)(Love, Huber, & Anders, 2014) with the cutoff values of Benjamini-Hochberg adjusted p-value < 0.05 and  $|\log_2(\text{fold change})| > 0.5$ . The gene expression count matrix was converted into the reads per kilobase per million mapped reads (RPKM) matrix, followed by Z score normalization for visualization.

## CUT&Tag and data processing

For the CUT&Tag experiments,  $8 \times 10^5$  cells were processed using the CUT&Tag Kit (Vazyme, TD903). The cells were collected and incubated with ConA beads and then mixed with the primary antibody pSmad1/5/9 (CST, #13820) overnight at 4°C. After incubation with the secondary antibody for one hour, the cells were further incubated with pA/G-Tnp for one hour. The transposase was then activated for DNA fragmentation. Following DNA extraction, library amplification and purification were performed. The libraries were sequenced using paired-end reads on the Illumina HiSeq PE150 platform.

For data processing, raw CUT&Tag data was trimmed using Trim Galore (version 0.6.10) software, and the clean reads were mapped to the human reference genome hg38 using Bowtie2 (version 2.5.1) software(Langmead & Salzberg, 2012) with the following parameters: "--very-sensitive --no-mixed --no-discordant -I 10 -X 1000". The SAM files were transformed into bam files via SAMtools (version 1.17) software(Li et al., 2009). The CUT&Tag peaks were performed using the MACS2 (version 2.2.8) with the default parameters(Zhang et al., 2008). Differential peaks were calculated using the R package DiffBind (version 3.4.11) with the cutoff values of adjusted p-value < 0.05 and  $|\log_2(\text{fold change})| > 0.5$  and annotated using the R package ChIPpeakAnno (version 3.28.0)(Lihua, Claude, & Nathan, 2010). To visualize the CUT&Tag signal, the reads per kilobase per million mapped reads (RPKM) were calculated in each 50-bp bin size using DeepTools (version 3.5.2) software(Ramirez, Dundar, Diehl, Gruning, & Manke, 2014), and the RPKM matrix was normalized using the Z score algorithm.

## ChIP-qPCR

At room temperature, approximately  $2 \times 10^6$  IMR90 cells were cross-linked with 1% paraformaldehyde for 10 minutes. The cross-linking reaction was stopped by adding 0.125 M glycine for 6 minutes. The cells were then harvested and lysed on ice in nuclear lysis buffer (50 mM Tris-Cl, 10 mM EDTA, 1% SDS, and protease inhibitors). Chromatin immunoprecipitation (ChIP) was performed using a Bioruptor (Diagenode) to shear the chromatin. Fragmented chromatin (25  $\mu$ g) was immunoprecipitated with 5  $\mu$ g of anti-pSmad1/5/9 antibody (CST, #13820) or anti-rabbit IgG (Abcam, ab171870). The immunoprecipitated DNA was quantified via RT-qPCR. The amount of DNA in the immunoprecipitates was calculated as a percentage relative to 10% of the input chromatin. All ChIP-qPCR-related primers are available in Table S2.

## Immunofluorescence

The slides were washed twice in PBS, fixed with 4% PFA for 10 minutes, and washed again three times with PBS. Subsequently, permeabilization was performed via incubation with 0.5% Triton X-100 for 10 minutes, followed by three additional washes with PBS. Next, blocking was performed with 1% BSA, and after blocking, the slides were washed three times with PBS. The primary antibody was then diluted at a ratio of 1:500 in PBST (PBS + 0.1% Tween 20) and incubated overnight at 4°C. Afterward, the slides were washed four times with PBST, with each wash lasting 5 minutes. The secondary antibody was diluted at a ratio of 1:500 in PBST (1  $\times$  PBS + 0.1% Triton X-100) and incubated at room temperature for 1–1.5 hours. After three washes with PBST, a final wash with PBS was conducted. DAPI was diluted to a final concentration of 1–2 ng/ $\mu$ L and incubated at room temperature for 3–5 minutes. Following three washes with PBS and one wash with ddH<sub>2</sub>O, the slides were sealed with Fluoromount-G<sup>®</sup> (SouthernBiotech, 0100-010) and covered with a coverslip.

## Lentiviral production and viral transduction

First, the pLKO.1 empty vector or vectors encoding shRNAs targeting *ID1* or *ID2* (Table S3), along with the assistant plasmids psPAX2 and pMD2.G, were cotransfected into HEK293T cells. After 18 h of transfection, the medium was replaced with DMEM supplemented with 30% complement-inactivated FBS, and viral supernatants were collected after 24 h and 48 h of transfection. When IMR90 cells reached 70% confluence, they were infected with the packaged virus with 5  $\mu$ g/mL polybrene (MCE, HY-K1057). At 36–48 hours after infection, successfully transduced cells were selected via treatment with 1  $\mu$ g/mL puromycin (M&C Gene Technology, MA009) for 24–36 hours.

## Detection of Reactive Oxygen Species (ROS)

IMR90 cells were harvested from a 6-well plate. After the cells were washed, the ROS-sensitive fluorescent probe 2',7'-dichlorofluorescein diacetate (DCFH-DA, MCE, HY-D0940) was used for detection. A 10 mM stock solution of DCFH-DA was prepared by dissolving it in anhydrous DMSO. This stock solution was subsequently diluted in serum-free DMEM to obtain a 10  $\mu$ M working solution of DCFH-DA. The samples were incubated with the DCFH-DA probe at 37°C for 30 minutes. After incubation, the samples were washed to remove excess probe. The fluorescence intensity was measured via a fluorescence microscope. The fluorescence intensity corresponds to the level of ROS present in the sample.

## Mice feeding

All the mouse experiments were approved by the Institutional Animal Care and Use Committee of Peking University. Twelve-month-old C57BL/6 mice (purchased from Wukong Biotech Co., Ltd.) were housed in a specific pathogen-free (SPF) animal facility under controlled environmental

conditions with a temperature range of 20–25°C, 55±10% humidity, and a 12-h light/dark cycle. The mice had ad libitum access to food and water. A daily dosage of 30 mg/kg of enalapril was dissolved into the mice's drinking water. Four to ten mice were included in each group.

### Forelimb grip strength test

The mice were placed on a grip strength meter, and their tails were gently pulled to allow them to grasp the pull bar with their forelimbs. The grip strength meter readings were recorded at the point of maximum force exerted by each mouse. The measurements were repeated multiple times to ensure reliable and consistent results.

### Rotarod fatigue test

The mouse was placed on the rotarod, and rotation was initiated. The device was set to an accelerating mode that increased from 4 to 40 revolutions per minute over 300 seconds. The time at which each mouse fell off or lost balance was recorded. The test was repeated multiple times to assess the endurance and motor performance of the mice.

### Y-maze test

The Y-maze consists of three opaque arms arranged at 120° angles relative to each other. The mice were placed in the central position of the maze and allowed to explore the three arms freely for 10 minutes. Each consecutive entry into one of the three arms was defined as spontaneous alternation behavior. The total number of arm entries and the number of spontaneous alternations were recorded. The percentage of spontaneous alternation behavior was then calculated.

### Open field test

The mice were removed from their cages and placed in an open field, with the central area designated. The mice were allowed to move freely within the open field for 10 minutes, and the number of times that they crossed the central area was recorded.

### Mouse serum extraction and analysis

After the mice were anesthetized, blood (approximately 1 mL per mouse) was collected from the eyeballs. The blood was allowed to clot and settle at room temperature. The mixture was then centrifuged at 3,000 rpm for 20 minutes, and the upper clear layer, which was the serum, was collected. The serum was subsequently subjected to biochemical assays, including ALT, CREA, LDH, TG, LDL, and HDL tests.

### Mouse tissue extraction

After the mice were euthanized, they were perfused with prechilled PBS until no blood was expelled. The heart, liver, spleen, kidney, brain, and muscle tissues were harvested. A portion of each tissue sample was used for RNA and protein extraction for sequencing and Western blot analysis, respectively, while another portion was fixed in 4% paraformaldehyde (PFA). The tissues were then subjected to either frozen or paraffin embedding. Various staining techniques, including  $\beta$ -galactosidase ( $\beta$ -gal) staining, periodic acid-Schiff (PAS) staining, Congo red staining, Sirius red staining, Oil Red O staining, and hematoxylin and eosin (H&E) staining, were performed.

### Pathway enrichment analysis

Gene Ontology (GO) enrichment analysis was performed with Metascape(Zhou et al., 2019 [4](#)), and gene set enrichment analysis (GSEA) was performed with the R package ClusterProfiler (version 4.2.2)(Wu et al., 2021 [5](#)). Pathways with p values < 0.05 were considered significantly enriched.



Representative terms were visualized with the ggplot2 (version 3.3.6) R package and ComplexHeatmap (version 2.10.0) R package (Gu, 2022 [↗](#)).

## Statistical analysis

The data are expressed as means  $\pm$  SEMs. The data were analyzed via a two-tailed unpaired Student's t test. A p value  $< 0.05$  was considered statistically significant.

## Results

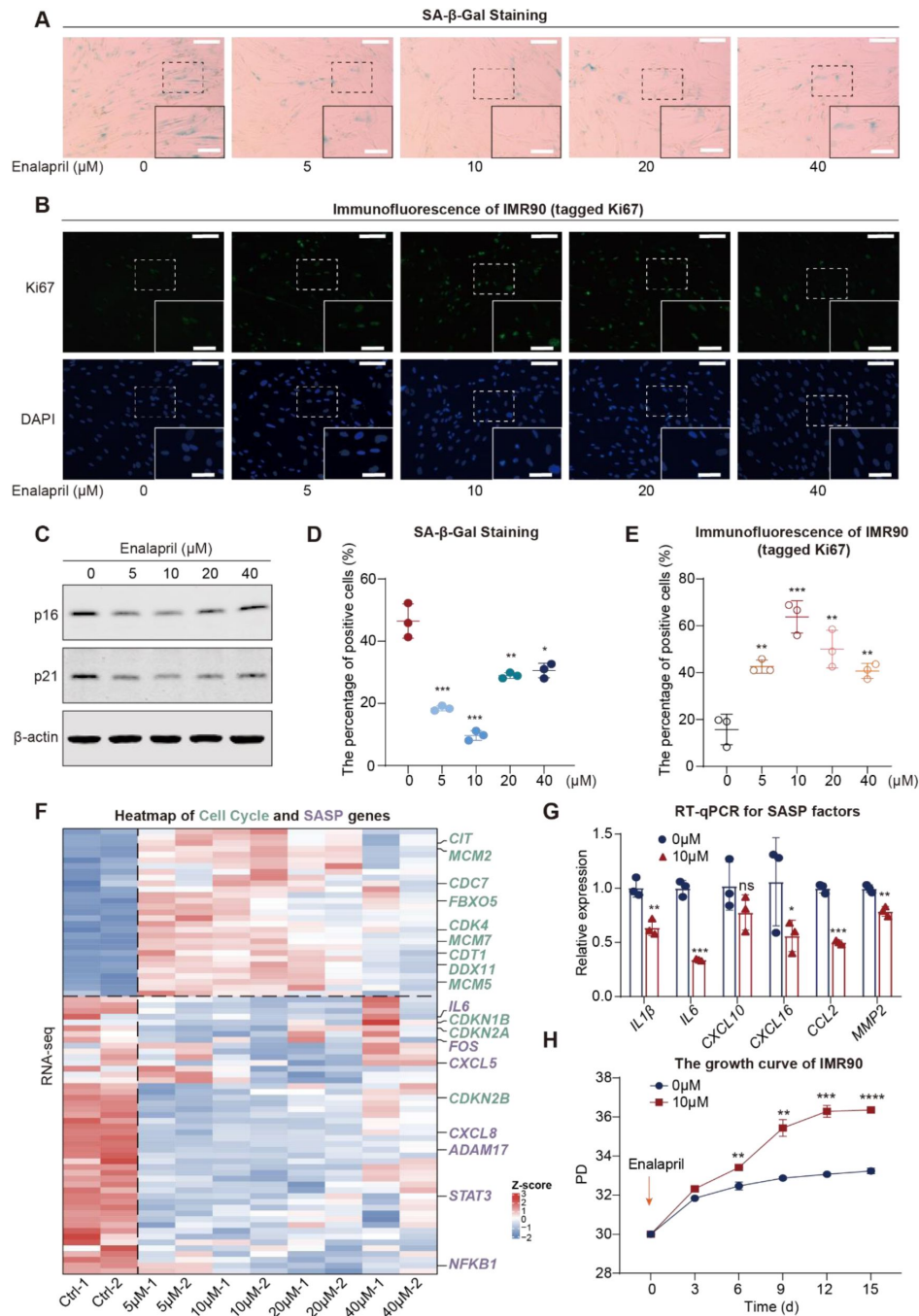
### Enalapril restrains cellular replicative senescence

To identify drugs that are capable of delaying cellular senescence, we established an in vitro replicative senescence model using primary human embryonic lung fibroblasts (IMR90). Compared with young cells, long-term-cultured IMR90 cells presented a characteristic senescence phenotype, characterized by increased SA- $\beta$ -gal staining, increased expression of the cell cycle arrest factors p16 and p21, and reduced expression of the proliferation marker Ki67 (Fig. S1A-C). Additionally, the release of SASP factors that significantly alter the cellular microenvironment, including the well-described factors *IL-1 $\beta$* , *IL-6*, and *CXCL10*, has also been reported to increase the degree of senescence (Fig. S1D).

Employing a drug repurposing strategy, we discovered that enalapril, an antihypertensive medication, delayed senescence. Notably, enalapril treatment at various concentrations reduced SA- $\beta$ -gal staining and increased Ki67 positivity in IMR90 cells (Fig. 1A-B, D-E [↗](#)). Western blot analysis confirmed a decrease in the aging markers p16 and p21 with the optimal effect of 10  $\mu$ M enalapril (Fig. 1C [↗](#)). To further analyze the effects of enalapril on transcriptome dynamics upon senescence entry, RNA-seq was performed across enalapril-treated cells at different concentrations. In general, we found that the expression of SASP genes decreased significantly (Fig. 1F [↗](#)). The expression of cell cycle arrest factors typically increased during senescence; however, the expression of these factors decreased after exposure to enalapril, whereas the expression of cell cycle genes related to cell proliferation increased upon treatment with enalapril at the optimal concentration of 10  $\mu$ M (Fig. 1F [↗](#)). We therefore treated the cells with 10  $\mu$ M enalapril via RT-qPCR and detected that many prominent SASP genes, such as *IL-1 $\beta$*  and *IL-6*, were significantly reduced compared with those in the control (Fig. 1G [↗](#)). Moreover, enalapril treatment increased the cell growth rate and delayed the transition to the growth arrest phase (Fig. 1H [↗](#)). Our comprehensive analysis underscores enalapril's potential as an anti-senescence agent, warranting further exploration for its role in combating cellular senescence.

### The function of enalapril depends on phosphorylated Smad1/5/9

Given the potential of enalapril to decelerate senescence, we screened several important targets of the aging-related pathway to identify the underlying mechanism of senescence entry. We found that enalapril notably elevated the phosphorylation level of Smad1/5/9 without affecting Smad2/3 phosphorylation, suggesting a possible role for the BMP-pSmad1/5/9 pathway (Fig. 2A [↗](#), Fig. S2A). To validate the pathway alterations in the gene expression profile, GSEA confirmed that enalapril treatment increased BMP signaling (Fig. 2B [↗](#)). Concurrently, the level of the Smad4 protein, which is required for pSmad1/5/9 to enter the nucleus, also increased, suggesting that enalapril might facilitate the nuclear entry of pSmad1/5/9 and modulate downstream genes (Fig. 2A [↗](#)) (Guglielmi et al., 2021 [↗](#); Senft, Bikoff, Robertson, & Costello, 2019 [↗](#)). To further assess the impact on gene regulation, we employed CUT&Tag analysis of pSmad1/5/9 and found that enalapril treatment increased the pSmad1/5/9 binding signals at the transcription start site (TSS) regions (Fig. 2C [↗](#)).

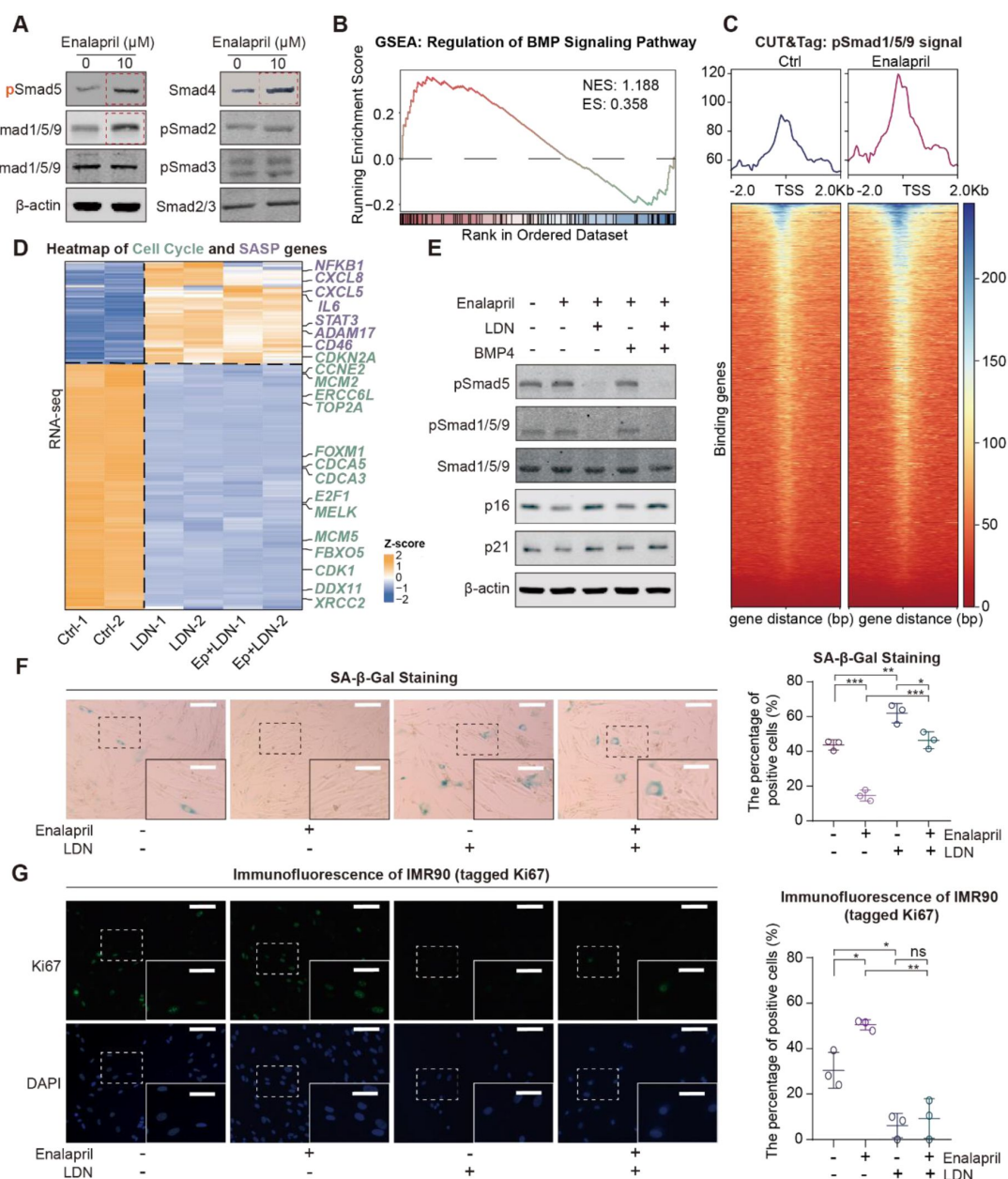


**Figure 1**

### Enalapril alleviates cellular senescence.

(A, D) SA-β-Gal staining (A) and statistical analysis of SA-β-Gal ratios (D) in IMR90 cells after enalapril treatment. Scale bars, 200 μm. Enlarged scale bars, 100 μm. (B, E) Ki67 immunofluorescence experiment (B) and statistical analysis of Ki67 intensity (E) after enalapril treatment. Scale bars, 80 μm. Enlarged scale bars, 40 μm. (C) Western blot analysis showing the protein levels of p16 and p21 in IMR90 cells after enalapril treatment. (F) Heatmap showing cell cycle-related gene (green) upregulation and SASP gene (purple) downregulation after enalapril treatment. (G) RNA expression of SASP factors in IMR90 cells after enalapril treatment, with blue bars representing the control group and red bars representing the enalapril-treated group. (H) Growth curves of IMR90 cells after enalapril treatment. The x-axis represents the passage time, while the y-axis represents population doubling (PD). A two-tailed t test was employed; ns indicates no significant difference, \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , \*\*\*\* $p < 0.0001$ .





**Figure 2**

### Phosphorylated Smad1/5/9 mediate the anti-senescence effect of enalapril.

(A) Western blot results for the detection of pSmad5, pSmad1/5/9 and other proteins after enalapril treatment. (B) GSEA showing the enrichment of the BMP signaling pathway after enalapril treatment. NES, normalized enrichment score. ES, enrichment score. (C) Profile of pSmad1/5/9 enrichment at the TSS regions in the control group (Ctrl) and the enalapril-treated group (Enalapril). (D) Heatmap of cell cycle (green) and SASP-related genes (purple) in the control group (Ctrl), BMP receptor inhibitor-treated group (LDN193189, LDN), and enalapril and BMP receptor inhibitor-cotreated group (EP+LDN). (E) Western blot analysis showing the protein levels of pSmad1/5/9, p16, and p21 in response to different combinations of enalapril, a BMP receptor inhibitor (LDN193189, LDN), and BMP4. (F) SA-β-Gal staining (left) and corresponding ratio statistical chart (right) of IMR90 cells treated with different combinations of enalapril, a BMP receptor inhibitor (LDN193189, LDN), and BMP4. Scale bars, 200 μm. Enlarged scale bars, 100 μm. (G) Ki67 immunofluorescence experiment (left) and intensity statistical chart (right) of IMR90 cells treated with different combinations of enalapril, a BMP receptor inhibitor (LDN193189, LDN), and BMP4. Scale bars, 80 μm. Enlarged scale bars, 40 μm. A two-tailed t test was employed; ns indicates no significant difference, \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

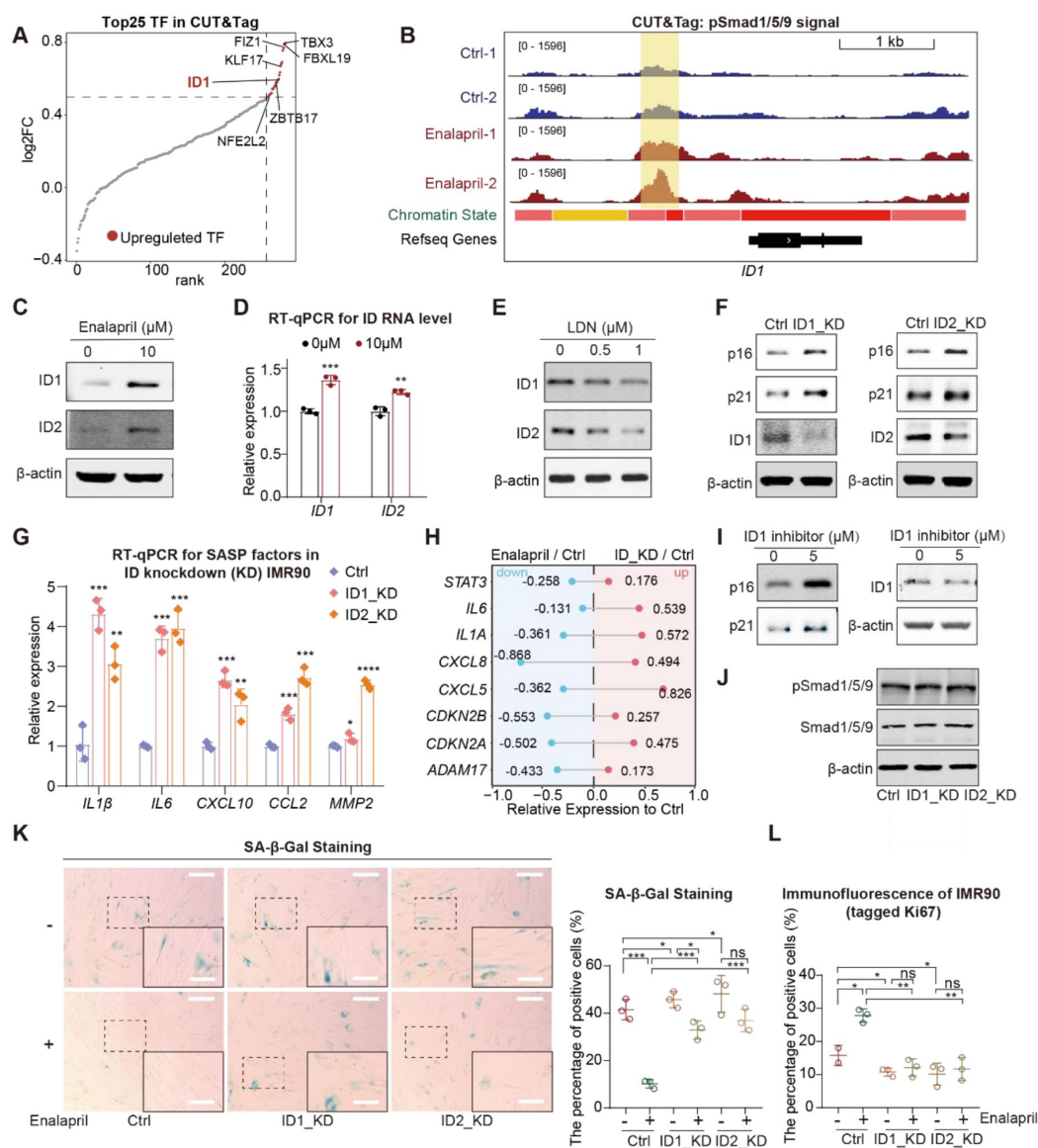
To explore the potential role of pSmad1/5/9 in anti-senescence mechanisms, we experimentally manipulated its protein levels by supplementing with BMP4 to increase pSmad1/5/9 levels or employing the BMP receptor inhibitor LDN193189 (LDN) to reduce it. Our findings revealed that BMP4 increased pSmad5 levels, which correlated with decreased p16 and p21, reduced  $\beta$ -gal staining and increased Ki67 positivity, indicating a mitigated senescent phenotype (Fig. S2B-D). In contrast, when cells were treated with the BMP receptor inhibitor, the changes in pSmad1/5/9 levels and senescence markers were completely reversed, highlighting the accelerated progression of senescence under the influence of the BMP receptor inhibitor (Fig. S2E-G).

Encouraged by the association between increased pSmad1/5/9, we tested whether pSmad1/5/9 serves as an intermediary in the anti-senescence effects of enalapril by combining LDN with enalapril. RNA-seq analysis revealed that LDN upregulated SASP genes and downregulated cell cycle genes. On the basis of these findings, the subsequent addition of enalapril did not reverse these changes, which suggests that the anti-senescence function of enalapril relies on pSmad1/5/9 (Fig. 2D). Furthermore, western blot results also revealed that in the presence of LDN, neither enalapril nor BMP4 restored Smad1/5/9 phosphorylation or the expression of p16 and p21 (Fig. 2E). Similarly,  $\beta$ -gal and Ki67 staining also revealed that enalapril failed to counteract LDN-induced deterioration (Fig. 2F, G). Collectively, these results indicate that elevated pSmad1/5/9 could increase the expression of critical genes, thereby exerting a potent anti-senescence effect.

## pSmad1/5/9 modulates the cell cycle and SASP secretion by upregulating ID1

The ID family, which is recognized as a key downstream target of pSmad1/5/9, serves as a biomarker for Smad1/5/9 phosphorylation levels (Genander et al., 2014; Hayashi et al., 2016; Ramachandran et al., 2018; Sun et al., 2022; Ying, Nichols, Chambers, & Smith, 2003). Coincidentally, we found that ID1 was the top-ranked transcription factor that pSmad1/5/9 bound to its promoter region via CUT&Tag analysis after enalapril treatment (Fig. 3A). Moreover, IGV analysis confirmed that enalapril significantly increased pSmad1/5/9 enrichment at the *ID1* promoter (Fig. 3B). Simultaneously, the expression levels of both *ID1* and *ID2* were elevated (Fig. 3C, D). Conversely, inhibiting pSmad1/5/9 with LDN led to decreased *ID1* and *ID2* levels, confirming the regulatory link between pSmad1/5/9 and the ID family (Fig. 3E). This finding indicated that the ID family was indeed a downstream gene regulated by pSmad1/5/9, and the increase in ID1 induced by enalapril further confirmed that enalapril could upregulate pSmad1/5/9 levels, thereby affecting the expression of downstream genes.

Prior studies have indicated that the ID family can suppress p16 and p21 expression, thus inhibiting cell cycle arrest and delaying cellular senescence (Hayashi et al., 2016; Lyden et al., 1999; Ying et al., 2003). This prompted us to examine ID protein expression during senescence, where we observed a decrease in ID1 and ID2 levels (Fig. S3A). The knockdown of *ID1* and *ID2* via shRNA or an ID1 inhibitor resulted in a decreasing trend of ID1 or ID2 expression and a significant increase in p16 and p21 expression (Fig. 3F, I). In addition, while  $\beta$ -gal increased the proportion of positively stained cells, the Ki67 proliferation signal decreased nearly fourfold after the knockdown of *ID1* or *ID2*, resembling the changes observed during senescence (Fig. S3B, C). Given that enalapril reduced SASP factors while LDN increased them, we investigated whether ID1, which acts as a transcriptional repressor downstream of pSmad1/5/9, could modulate SASP expression. Indeed, the expression of traditional SASP factors, such as *IL1 $\beta$*  and *IL6*, which decrease with enalapril treatment, significantly increased after *ID* knockdown (Fig. 3G). By overlapping the genes downregulated by enalapril with those upregulated upon *ID* knockdown, several SASP genes, as well as *CDKN2A* and *CDKN2B*, were identified (Fig. 3H). Consistent with these findings, we found that SASP factor expression also increased in *ID* knockout mouse bone marrow cells in previously reported data (Fig. S3E) (Fei et al., 2023). Importantly, *ID* knockdown did not affect pSmad1/5/9 levels, confirming that ID is a downstream effector of pSmad1/5/9, influencing the expression of genes such as p16, p21, and SASP-related genes (Fig. 3J).



**Figure 3**

### pSmad1/5/9 regulate ID-induced inhibition of p16, p21 and SASP.

(A) Changes in the peaks of all transcription factors identified by CUT&Tag following enalapril treatment, with red dots indicating transcription factors with increased peak intensity (Upregulated TFs). (B) Integrative Genomics Viewer (IGV) showing pSmad1/5/9 signals near the *ID1* region between the control group (Ctrl) and the enalapril treatment group (Enalapril). The vertical yellow boxes indicate regions with increased signal intensity. (C) Western blot analysis showing the changes in the protein levels of ID1 and ID2 following enalapril treatment. (D) Changes in the RNA levels of ID1 and ID2 following enalapril treatment. (E) Western blot analysis showing the protein levels of ID1 and ID2 following treatment with a BMP receptor inhibitor (LDN193189, LDN). (F, I) Western blot showing the changes in the protein levels of p16 and p21 following the knockdown of *ID1* or *ID2* (F) or the inhibition of *ID1* (I). (G) RNA expression of SASP factors after *ID* knockdown, with pink representing *ID1* knockdown and orange representing *ID2* knockdown. (H) Normalized average RNA expression levels of selected SASP factors and cell cycle arrest factors after enalapril treatment and *ID* knockdown. Positive values indicate upregulation, while negative values indicate downregulation. The values represent the expression levels relative to those of the Ctrl. (J) Western blot analysis showing pSmad1/5/9 levels following *ID1* and *ID2* knockdown. (K) SA-β-Gal staining (left) and SA-β-Gal ratio quantification (right) in the control group (Ctrl) and *ID*-knockdown groups (*ID1\_KD*, *ID2\_KD*) with or without enalapril treatment. Scale bars, 200 μm. Enlarged scale bars, 100 μm. (L) Ki67 immunofluorescence intensity in the control group (Ctrl) and *ID* knockdown groups (*ID1\_KD*, *ID2\_KD*), with or without enalapril treatment. A two-tailed t test was employed; ns indicates no significant difference, \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

After validating the prosenescence effects of *ID1* or *ID2* through *ID1* knockdown or the use of small molecule inhibitors, we next examined whether the anti-senescence effects of enalapril are mediated by ID proteins. The continued addition of enalapril to these *ID*-knockdown cells did not significantly alter  $\beta$ -gal or Ki67 signals compared with those in the *ID*-knockdown group without enalapril, implying that the benefits of enalapril may be ID dependent (Fig. 3K-L, Fig. S3D). Taken together, these findings indicate that enalapril elevates pSmad1/5/9 levels, which in turn upregulate *ID1* expression. This dual action has two key implications: first, it confirms that enalapril can modulate the expression of pSmad1/5/9 downstream genes. Second, by upregulating *ID1*, enalapril can suppress the expression of p16, p21, and SASP-related genes, thereby potentially restoring cell proliferation and reducing the expression of inflammatory factors. These findings suggest that the anti-senescence mechanism of enalapril involves the pSmad1/5/9-*ID1* axis, highlighting a novel pathway for its therapeutic effects on aging.

## Enalapril has antioxidative effects through pSmad1/5/9

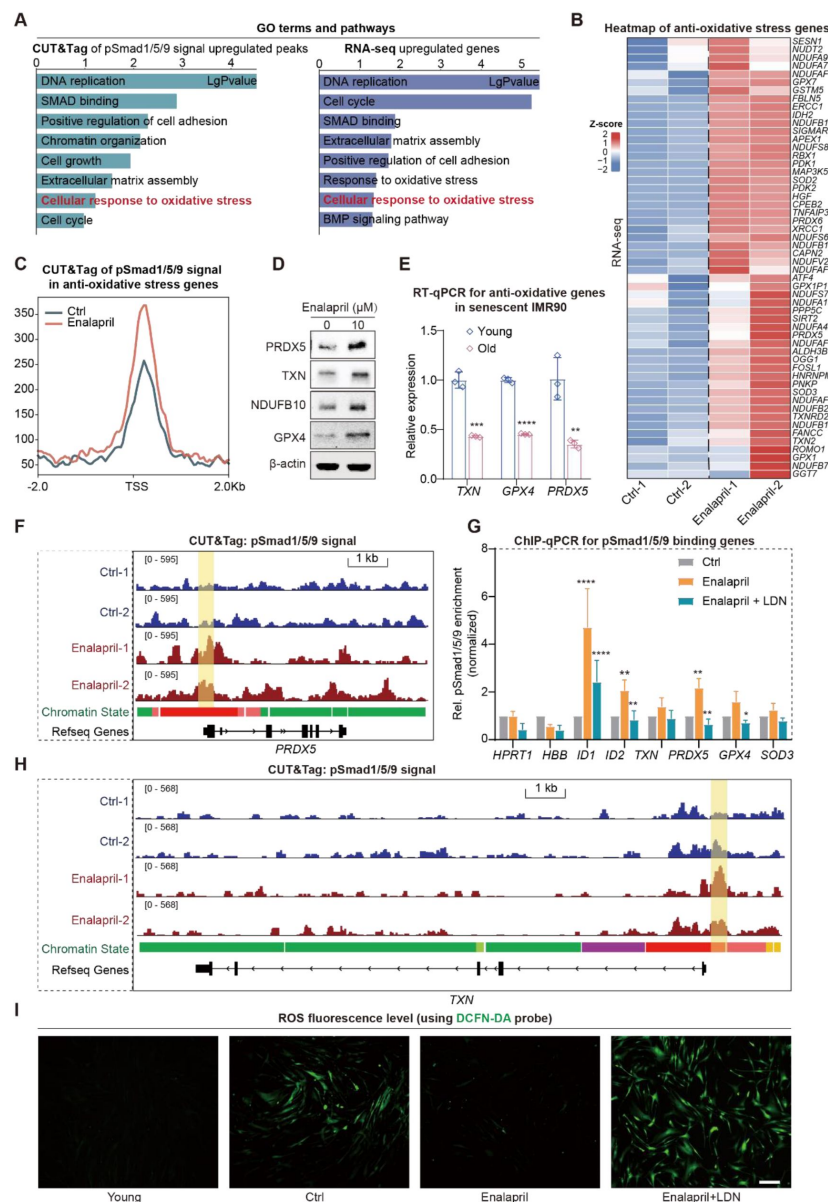
Since an increase in pSmad1/5/9 after enalapril treatment was observed and pSmad1/5/9 is a transcription factor, to understand the in-depth mechanism of enalapril regulation, we further explored the downstream genes of pSmad1/5/9. Analysis of CUT&Tag-upregulated peaks after enalapril treatment revealed that enalapril treatment led to enrichment of pSmad1/5/9 in the GO pathway associated with DNA replication, the cell cycle and the oxidative stress response, all of which are crucial for cell growth (Fig. 4A). Moreover, transcriptome analysis revealed that the genes whose expression increased after enalapril treatment were also enriched in pathways related to the response to oxidative stress (Fig. 4A). Notably, genes upregulated by enalapril in response to oxidative stress included antioxidative enzymes, electron transport chain components and lipid oxidation inhibitors, with a significant increase in pSmad1/5/9 at their TSS regions (Fig. 4B, C). Specifically, we also examined the promoter regions of antioxidative genes, such as the antioxidative enzyme *PRDX5* and the lipid oxidation inhibitor *TXN*, and found increased pSmad1/5/9 enrichment along with upregulated gene expression (Fig. 4D, F-H). During cellular senescence, a decrease in these antioxidative genes exacerbates oxidative stress, reduces energy metabolism, and increases lipid oxidation, leading to the release of inflammatory cytokines (Fig. 4E)(Coppe et al., 2008).

It has been reported that changes in the expression of antioxidative genes can affect the ROS level of cells; thus, we then used DCFH-DA probes to detect ROS levels in cells(Coppe et al., 2008). The results revealed that cellular senescence increased ROS, whereas enalapril slowed the increase in ROS, indicating its ability to upregulate antioxidative genes and reduce ROS, thus affecting cellular senescence (Fig. 4I). Importantly, the continued addition of BMP receptor inhibitors reversed the beneficial effects of enalapril on ROS and decreased pSmad1/5/9 enrichment of these antioxidative genes (Fig. 4G, I). Collectively, these results reinforce the notion that enalapril promotes the upregulation of antioxidative genes through pSmad1/5/9, reduces ROS levels and alleviates oxidative stress, thus playing a crucial role in delaying senescence.

## Enalapril upregulates pSmad1/5/9 and improves physiological function in mice

Enalapril, which is known for its anti-senescence potential, was further investigated for its impact at the organ and individual levels in 12-month-old mice (Fig. 5A). After continuous feeding at a dosage of 30 mg/kg/day for 3 months, we conducted a comprehensive RNA-seq analysis of six key organs—brain, kidney, liver, heart, muscle, and spleen—to assess the impact of enalapril on the transcriptome. We found that enalapril induced organ-specific gene expression changes, thereby affecting the physiological functions of these genes. Moreover, GSEA revealed upregulated pathways related to forebrain and kidney development, long-chain fatty acid metabolism and other processes, indicating that enalapril has a beneficial effect on organ functions (Fig. S4A). Additionally, enalapril reduced the expression of SASP factors in multiple organs, particularly the





**Figure 4**

### pSmad1/5/9 increase the expression of antioxidative genes.

(A) Bar plot showing the enriched GO terms and pathways associated with the CUT&Tag-upregulated peaks (left) and the upregulated RNA-seq genes (right) following enalapril treatment. (B) Heatmap showing the changes in the RNA expression of antioxidative genes. (C) Profile plot showing the increase in the pSmad1/5/9 binding signal of antioxidative genes in the TSS region following enalapril treatment. (D) Western blot analysis showing the protein levels of representative antioxidative genes after enalapril treatment. (E) Relative RNA levels of *TXN*, *GPX4*, and *PRDX5* during cellular senescence. The blue bars represent young cells, and the red bars represent senescent cells. (F, H) Integrative Genomics Viewer (IGV) showing pSmad1/5/9 signals near the *PRDX5* (F) and *TXN* (H) regions between the control group (Ctrl) and the enalapril treatment group (Enalapril). The vertical yellow boxes indicate regions with increased signal intensity. (G) ChIP-qPCR results showing pSmad1/5/9 levels at many peaks following enalapril or combination treatment with enalapril and LDN193189. The y-axis represents the normalized pSmad1/5/9 signals relative to 10% input. pSmad1/5/9 enrichment for *HPRT1* and *HBB* served as a negative control, and pSmad1/5/9 enrichment for *ID1* and *ID2* served as positive controls, as previously described. A two-tailed t test was employed, \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , \*\*\*\* $p < 0.0001$ . (I) Detection of ROS fluorescence levels via the DCFH-DA probe in young cells, senescent cells (Ctrl), cells treated with enalapril, and cells treated with a combination of enalapril and LDN193189. Scale bars, 200  $\mu\text{m}$ .

CCL, CXCL, and MMP families (Fig. 5B). Serum SASP array analysis also revealed a decrease in the secretion of certain SASP proteins, such as CCL6, CCL11, CXCL13, IGFBP-1, and MMP3, suggesting that enalapril has anti-inflammatory effects in mice (Fig. 5C).

Given the known upregulation of pSmad1/5/9 levels and antioxidative genes in cellular studies by enalapril, we measured Smad1/5/9 phosphorylation and the expression of the antioxidative genes *Txn1*, *Txn2*, and *Prdx5* in the tested organs. We observed increased pSmad1/5/9 levels, especially in the brain, kidney and liver, along with upregulated *Id1* and antioxidative genes, notably *Txn1* and *Txn2* (Fig. 5D, Fig. S4B). Further analysis of BMP signaling and antioxidative stress-related pathways in these organs revealed increased gene expression after enalapril treatment, indicating the activation of this pathway (Fig. 5E). These findings align with our cellular findings, implying that enalapril may enhance the pSmad1/5/9-antioxidative gene axis to increase organ physiological function and delay individual aging.

## Enalapril alleviates aging-related behavior in naturally aged mice

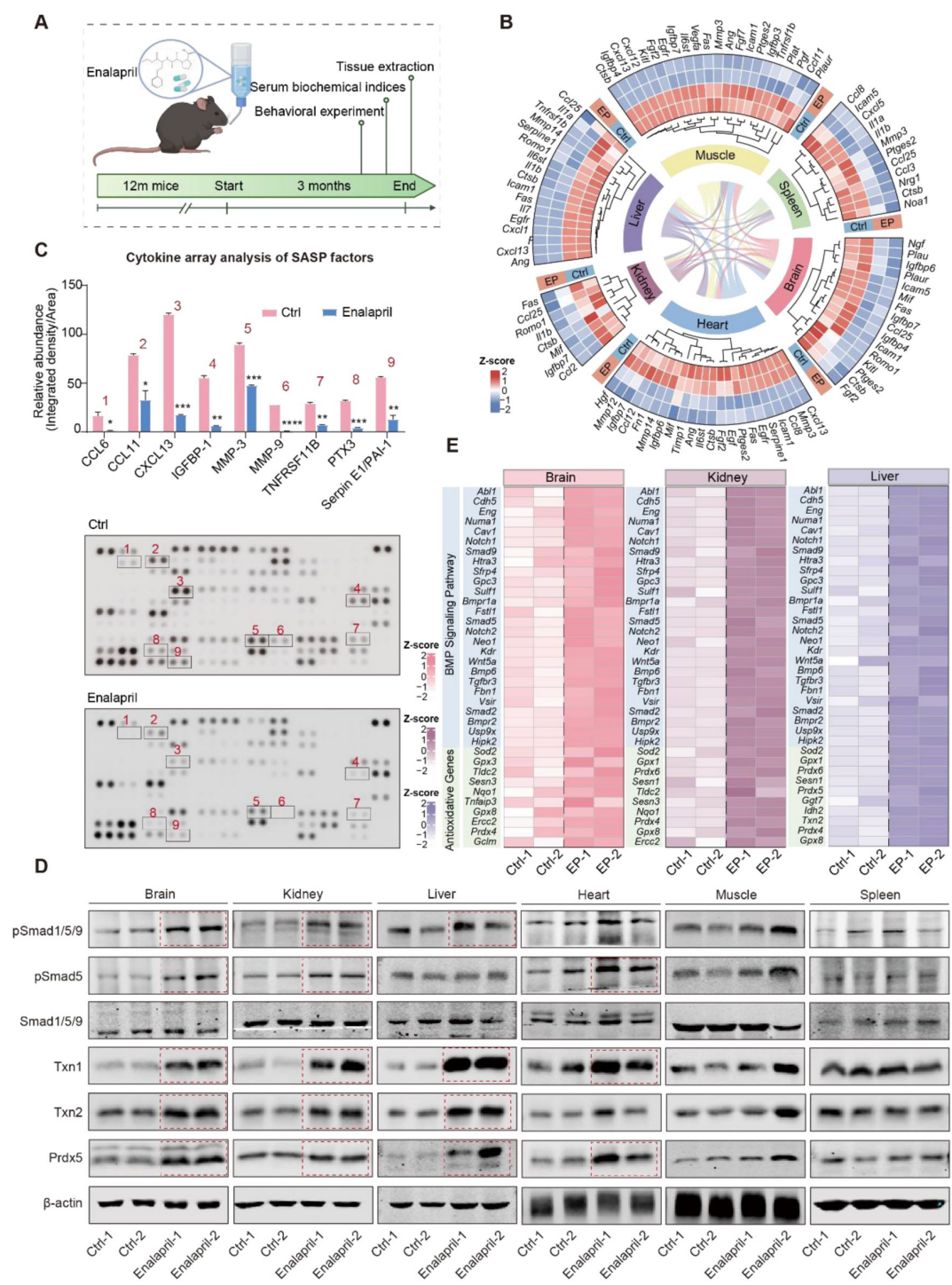
Building on our finding of pSmad1/5/9 activation and antioxidative genes upregulation in multiple organs, we further investigated the broader phenotypic effects of enalapril in mice. After three months of enalapril feeding, behavioral experiments, serum tests, and tissue section staining were performed. Notably, enalapril had no significant effect on body weight in the mice, suggesting that the drug has no overt toxic side effects (Fig. 6A). However, enalapril improved the decrease in age-related mobility, as evidenced by increased grip strength and extended time in the rotarod fatigue experiments, suggesting increased muscle strength and endurance (Fig. 6D, E).

During the aging process, the brain undergoes varying degrees of functional decline. To assess the impact of enalapril on brain function, we conducted Y-maze and open field tests. The Y-maze, which is frequently employed to assess short-term memory in mice by measuring spontaneous alternation as a marker of spatial working memory, indicates notable improvements. Under normal circumstances, mice prefer to explore new maze branches rather than return to previously visited branches. Several brain regions, including the hippocampus, septum, basal forebrain, and prefrontal cortex, contribute to this task, and the age-related decline in spontaneous alternation ability in mice is associated with degeneration or lesions in brain regions such as the hippocampus (Radulescu, Cerar, Haslehurst, Kopanitsa, & Barnes, 2021). Enalapril promoted a substantial increase in the rate of spontaneous alternation in the mice, indicating enhanced spatial memory (Fig. 6B, C). As aging progresses, anxiety-like behaviors often increase with age in mice (Radulescu et al., 2021). The open-field test, a valuable tool for assessing autonomous and exploratory behaviors in new environments, demonstrated that enalapril increased the number of central area crossings, suggesting a reduction in aging-related anxiety (Fig. 6F, G). In conclusion, these results suggest that enalapril has the potential to mitigate the deterioration of various behavioral aspects associated with aging, including endurance, spatial memory and anxiety.

## Enalapril ameliorates aging-related pathological changes in aged mice

The results of these behavioral tests provided valuable insights. To further explore the effects of enalapril on pathological changes, our investigation was extended to serum analyses. This comprehensive analysis encompassed the evaluation of vital organ function, including the liver, kidneys, and heart, as well as an assessment of lipid content, specifically triglycerides (TGs), low-density lipoprotein cholesterol (LDL), and high-density lipoprotein cholesterol (HDL). After enalapril administration, there was no significant change in the serum alanine transaminase (ALT) level, indicating that no liver toxicity from enalapril occurred (Fig. 6H). Moreover, the level of creatinine (CREA), a key marker of kidney function, was significantly decreased in enalapril-treated mice, indicating potential functional recovery and improvement of renal function (Fig. 6I). A heart health indicator, lactate dehydrogenase (LDH), was also reduced, indicating a

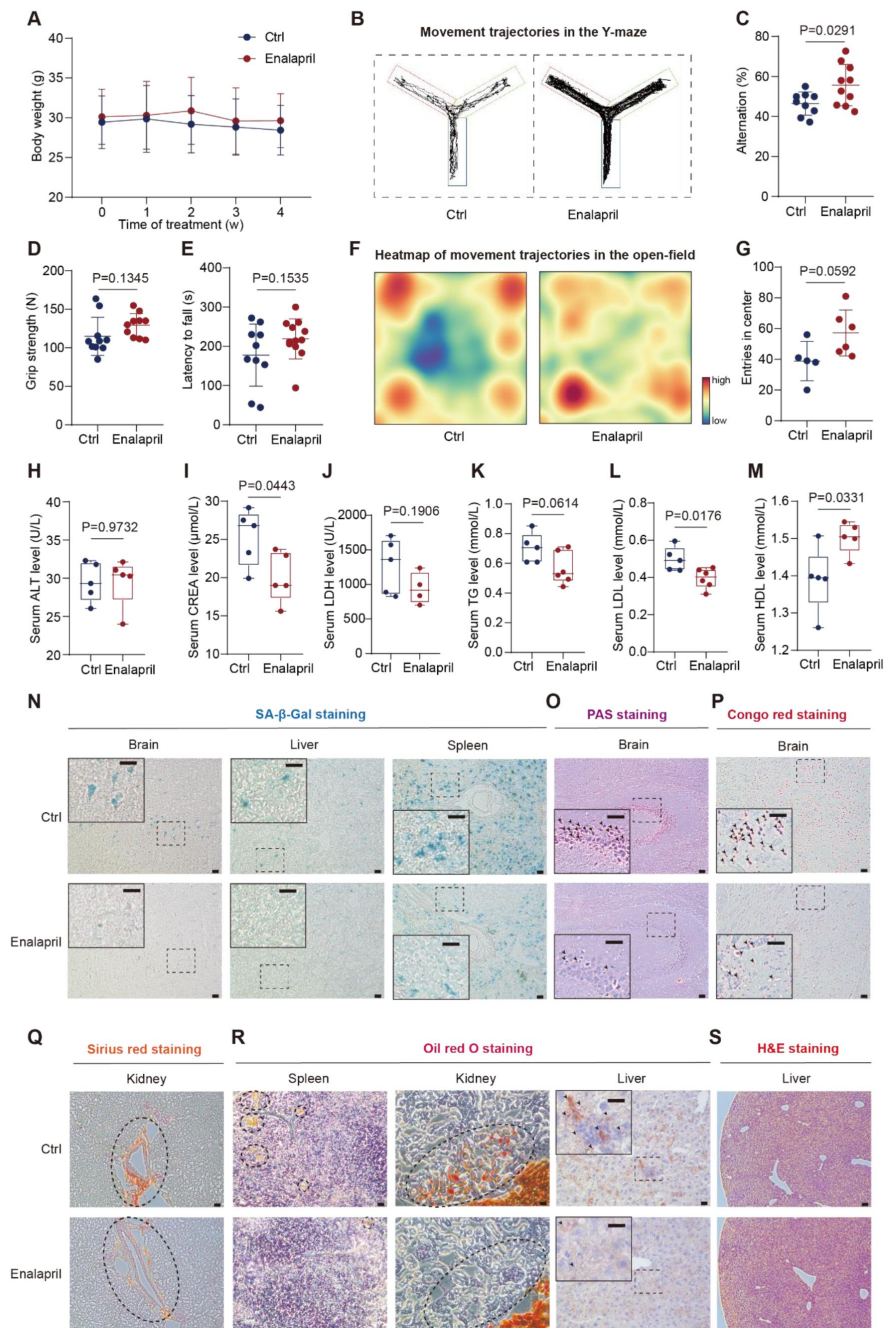




**Figure 5**

**Enalapril exerts its anti-aging effects on mouse organs via pSmad1/5/9-antioxidative genes.**

(A) A schematic workflow of the mouse feeding and experimental procedures. Created in BioRender. Li, D. (2024) BioRender.com/u88h339 (B) Heatmap of RNA levels of SASP factors in various organs following enalapril treatment (EP). (C) Cytokine array analysis of secreted proteins (below) and relative quantitation (above) of SASP factors in control serum (Ctrl) and enalapril-treated serum (Enalapril). (D) Western blot analysis showing the protein levels of pSmad1/5/9 and antioxidative genes in various organs following enalapril treatment. (E) Heatmap of the RNA levels of BMP signaling pathway-related genes and antioxidative genes in the brain, kidney, and liver following enalapril (EP) treatment.



**Figure 6**

### Enalapril mitigates age-related degenerative changes in aged mice.

(A) Changes in the body weights of the mice after enalapril feeding. (B, C) Movement trajectory (B) and spontaneous alternation rate (C) in the Y-maze of the mice after enalapril treatment. (D, E) Changes in the grip strength (D) and rotarod fall time (E) of the mice after enalapril treatment. (F, G) Heatmap of movement trajectories (F) and the number of entries into the central area (G) in the open-field test after enalapril treatment. (H–M) Alterations in the serum levels of aspartate transaminase (AST) (H), creatinine (CREA) (I), lactate dehydrogenase (LDH) (J), triglycerides (TGs) (K), low-density lipoprotein cholesterol (LDL) (L), and high-density lipoprotein cholesterol (HDL) (M) after enalapril treatment in mice. (N) SA-β-Gal staining of the brain, liver, and spleen after enalapril treatment. Scale bars, 20 μm. (O, P) Periodic Acid-Schiff (PAS) staining (O) and Congo red staining (P) of the brain after enalapril treatment. Scale bars, 50 μm. (Q) Sirius red staining of the kidney after enalapril treatment. Scale bars, 50 μm. (R) Oil Red O staining of the spleen, kidney and liver after enalapril treatment. Scale bars, 20 μm. (S) Hematoxylin and eosin (H&E) staining of the liver after enalapril treatment. Scale bars, 50 μm.

positive cardiac effect of enalapril (**Fig. 6J**). These findings underscore the multifaceted effects of enalapril on various physiological parameters, emphasizing its potential as a therapeutic intervention in aging-related conditions.

The transcriptomic data from the enalapril treatment group revealed an increase in long-chain fatty acid metabolism in the liver, suggesting that enalapril may play a positive role in lipid metabolism (**Fig. S4A**). Accordingly, we analyzed serum lipid levels in enalapril-treated mice. Typically, aging leads to a decline in lipid metabolism in the serum, causing an accumulation of triglycerides (TGs) and low-density lipoprotein (LDL) and a decrease in high-density lipoprotein (HDL), contributing to oxidative stress and conditions such as hyperlipidemia and fatty liver (A. A. Johnson & Stolzing, 2019). Following enalapril treatment, the mice presented significant reductions in TG and LDL levels and a noticeable increase in HDL levels (**Fig. 6K-M**). These results suggest that enalapril has the potential to increase lipid metabolism in aged mice, thereby mitigating the increase in blood lipid levels.

To evaluate the improvements in the aging process induced by enalapril in various organ systems, we conducted pathological examinations of mouse tissues. A decrease in  $\beta$ -gal staining was observed in the brain, liver and spleen of enalapril-treated mice, suggesting a delay in organ aging (**Fig. 6N**). Given the effects of enalapril on brain function, we further examined pathological markers in the brain. Periodic acid-Schiff (PAS) staining, which is indicative of glycogen deposition, revealed a significant reduction in glycogen near the hippocampus (**Fig. 6O**). Congo red staining, which detects amyloid protein deposition, also revealed reduced staining in the cerebral cortex (**Fig. 6P**). In the kidney, Sirius red staining, which highlights abnormal collagen fibers or fibrosis, revealed significant red staining around blood vessels in control mice. However, this staining was noticeably reduced in the kidneys of the mice treated with enalapril (**Fig. 6Q**). These findings suggest that enalapril has a beneficial effect on fibrosis in the kidneys of aged mice, which aligns with the observed improvement in plasma creatinine levels (**Fig. 6I**). Furthermore, analysis of spleen, kidney and liver sections from enalapril-treated mice revealed a reduction in Oil Red O staining, with the kidneys and liver showing the most significant changes (**Fig. 6R**). This finding aligned with the consistent decrease in lipid-related molecular content in plasma, indicating that enalapril has a role in reducing age-related lipid accumulation (**Fig. 6K-M**). Hematoxylin and eosin (H&E) staining of the liver revealed a decrease in cavities and increased tissue density in mice treated with enalapril, suggesting its role in enhancing liver cell proliferation and regeneration (**Fig. 6S**).

In summary, enalapril significantly mitigated multiple pathological markers associated with aging, including lipid droplets, amyloid, glycogen deposition, and fibrosis, in the brain, kidneys, liver, heart and spleen of aged mice. These comprehensive improvements contribute to an overall amelioration of their aging status.

## Discussion

Since the early 2000s, the National Institutes of Health (NIH) has been pioneering the Interventions Testing Program (ITP), an initiative aimed at harnessing existing drugs or small molecules to decelerate the aging process (Harrison et al., 2009; Nadon et al., 2008; Strong et al., 2016). This approach repurposes established medications for novel uses, maximizing the potential of known compounds. It has accelerated the transition of anti-aging therapies from discovery to clinical application, ensures their safety and bolsters a comprehensive anti-aging drug screening strategy.

In our study, we discovered that enalapril, a commonly prescribed ACE inhibitor for hypertension, has significant anti-aging effects at both the cellular and organismal levels (**Fig. 7**). Enalapril demonstrated a notable ability to attenuate aging-related physiological decline across multiple

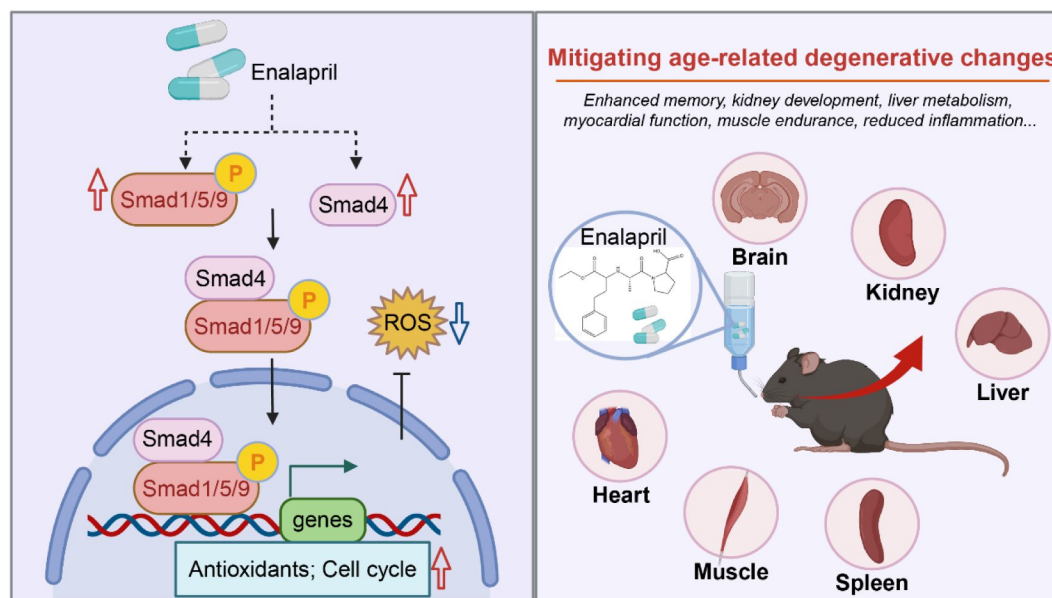


organs, including the brain, kidneys, and liver. This finding suggests its potential to reduce the risk of disease across multiple organs, slow the overall aging process and extend both the healthspan and lifespan of mice. This observation aligns with previous reports on the lifespan-extending effects of other ACE inhibitors, such as captopril, which have been shown to increase longevity in organisms such as *C. elegans* and mice (Egan et al., 2024 [↗](#); Kumar, Dietrich, & Kornfeld, 2016 [↗](#); Strong et al., 2022 [↗](#)). However, our screening process revealed that enalapril outperformed captopril and other antihypertensive drugs in ameliorating senescent phenotypes, although captopril and other antihypertensive drugs also exhibited the ability to delay senescence. These findings indicate that enalapril may have unique or more potent mechanisms of action that contribute to its anti-aging effects, warranting further mechanistic studies to better understand its molecular targets.

Mechanistically, our data revealed that enalapril enhances the phosphorylation of Smad1/5/9, a critical signaling pathway that regulates various cellular processes, including cell proliferation and the oxidative stress response. Phosphorylation of Smad1/5/9 facilitates its nuclear translocation, where it acts as a transcriptional regulator to upregulate downstream genes, such as those involved in cell cycle regulation and antioxidative responses. These findings suggest that the ability of enalapril to modulate the pSmad1/5/9 pathway contributes to its protective effects against oxidative stress, a well-established hallmark of aging. By reducing ROS levels and promoting cell proliferation, enalapril effectively delays the progression of cellular senescence and organismal aging. This finding not only highlights the potential of phosphorylated Smad1/5/9 as a therapeutic target for anti-aging interventions but also reveals a novel antioxidative mechanism of enalapril action.

Several FDA-approved drugs have been repurposed for anti-aging applications, demonstrating a wide range of mechanisms and effects. These include rapamycin, metformin, and NAD<sup>+</sup> precursors. For example, rapamycin, which was originally used as an immunosuppressant to prevent organ transplant rejection, inhibits the mTOR pathway, which is associated with cell growth and metabolism (S. C. Johnson, Rabinovitch, & Kaeberlein, 2013 [↗](#)). It extends the lifespan of mice and shows potential anti-aging effects across various species (Harrison et al., 2009 [↗](#)). Compared with other repurposed anti-aging drugs, enalapril offers distinct advantages because of its unique mechanism of action. As an ACE inhibitor, enalapril inherently targets the renin-angiotensin system, providing vascular protection (Schmieder, Hilgers, Schlaich, & Schmidt, 2007 [↗](#); Yamamoto & Rakugi, 2021 [↗](#)). Compared with repurposed drugs such as rapamycin, which may have more pronounced side effects, such as immunosuppression, limiting their long-term use, enalapril has a well-established safety profile (Janes & Fruman, 2009 [↗](#); Saunders, Metcalfe, & Nicholson, 2001 [↗](#)). This can facilitate faster clinical translation for anti-aging applications. Moreover, our study revealed that enalapril reduces ROS through the pSmad1/5/9-antioxidant axis, highlighting its potential for antioxidation, a hallmark of aging that is less directly targeted by drugs such as rapamycin or metformin. Additionally, enalapril enhances multi-organ function, particularly in the brain, kidneys, and liver. Its well-established safety profile further supports enalapril as a promising candidate for clinical anti-aging applications, with the potential to complement existing therapies by promoting both vascular health and systemic antioxidative and anti-inflammatory effects.

Aging is a highly complex and multifactorial process (Campisi, 2013 [↗](#); Jazwinski, 1996 [↗](#)). It involves intricate interactions among various signaling pathways and crosstalk between tissues and organs, especially during individual aging (Jazwinski, 1996 [↗](#); Munoz-Espin & Serrano, 2014 [↗](#)). Notably, enalapril is not uniform across all organs and tissues, as indicated by the differential improvement in aging phenotypes among various tissues. For example, enalapril significantly improved the physiological functions of the brain, muscle and other organs, but the specific pathways and mechanisms by which these effects are mediated might differ among organs. This observation suggests that the tissue-specific effects of enalapril may be mediated by differential activation of signaling pathways, including those beyond the Smad1/5/9 axis, which are known to



**Figure 7**

### Summary schematic.

It showing pSmad1/5/9-mediated combating cellular senescence of enalapril, and improvement of organ function in mice. Specifically, enalapril increases the level of pSmad1/5/9 and Smad4, promoting the expression of downstream genes, such as antioxidative and cell cycle related genes, thereby promoting cell proliferation and reducing ROS. Moreover, enalapril mitigates age related degenerative changes, including enhanced memory, improved kidney function, increased liver metabolism and other organ functions. Created in BioRender. Li, D. (2024) BioRender.com/u88h339.



play crucial roles in aging. Future studies should focus on delineating these tissue-specific mechanisms to fully understand the scope of the anti-aging effects of enalapril. Moreover, improvements in metabolic function in the liver could have downstream effects on brain function, as metabolic dysregulation is known to exacerbate neurodegenerative diseases. Similarly, enhanced vascular health resulting from the antihypertensive properties of enalapril may contribute to improved brain perfusion, reducing the risk of age-related cognitive decline. Therefore, exploring the systemic effects of enalapril and how improvements in one organ system may benefit others should be a priority for future research. Furthermore, given the established role of enalapril as an antihypertensive drug, assessing its potential side effects when it is repurposed for anti-aging purposes is essential. Identifying strategies to minimize these side effects while optimizing their anti-aging effects is an important area of investigation. In conclusion, our study identified enalapril as a promising candidate for repurposing as an anti-aging drug with antioxidative and promoting effects on cell proliferation. These findings open avenues for the development of targeted therapies aimed at mitigating aging and promoting healthy longevity in clinical settings, providing a foundation for guiding future research in the realm of anti-aging drug interventions.

## Data availability

All related sequencing data have been uploaded to the NCBI Gene Expression Omnibus and are accessible through accession numbers GSE277861 and GSE277862.

## Acknowledgements

We thank Dr. Xuehui Lyu (Peking University) and Shuo Chen (Center for Excellence in Molecular Cell Science, Chinese Academy of Sciences) for suggestions regarding data organization. We thank the National Center for Protein Sciences at Peking University for assisting with the cDNA library and the Core Facilities of Life Sciences, Peking University, for assisting with the confocal microscopy. We are also grateful to the Behavioral Platform of the Animal Center at Peking University for assisting with the behavioral experiments involving the mice.

## Additional information

### Author contributions

W.T. oversaw the study and obtained financial support. W.C.L. conducted the main bench experiments. H.C.W. analyzed the data. Z.H.D. and J.N.H. helped prepare the mice samples. R.W., F.J.M. and J.L.B. helped analyze the data. L.J.Z. helped guide the experiment. W.C.L., Y. T. G. and C. Z. prepared the original manuscripts. All the authors read and approved the final version.

### Funding

This work was supported by the National Key Research and Development Program of China (2021YFA0909300).

## References

- Baker D. J., Wijshake T., Tchkonia T., LeBrasseur N. K., Childs B. G., van de Sluis B., ..., van Deursen J. M. 2011) **Clearance of p16Ink4a-positive senescent cells delays ageing-associated disorders** *Nature* **479**:232–236 <https://doi.org/10.1038/nature10600>
- Bharath L. P., Agrawal M., McCambridge G., Nicholas D. A., Hasturk H., Liu J., ..., Nikolajczyk B. S. 2020) **Metformin Enhances Autophagy and Normalizes Mitochondrial Function to Alleviate Aging-Associated Inflammation** *Cell Metab* **32**:44–55 <https://doi.org/10.1016/j.cmet.2020.04.015>
- Campisi J 2013) **Aging, cellular senescence, and cancer** *Annu Rev Physiol* **75**:685–705 <https://doi.org/10.1146/annurev-physiol-030212-183653>
- Childs B. G., Gluscevic M., Baker D. J., Laberge R. M., Marquess D., Dananberg J., van Deursen J. M. 2017) **Senescent cells: an emerging target for diseases of ageing** *Nat Rev Drug Discov* **16**:718–735 <https://doi.org/10.1038/nrd.2017.116>
- Cohen A. A., Ferrucci L., Fülöp T., Gravel D., Hao N., Kriete A., ..., Varadhan R. 2022) **A complex systems approach to aging biology** *Nature Aging* **2**:580–591 <https://doi.org/10.1038/s43587-022-00252-6>
- Coppe J. P., Patil C. K., Rodier F., Sun Y., Munoz D. P., Goldstein J., ..., Campisi J. 2008) **Senescence-associated secretory phenotypes reveal cell-nonautonomous functions of oncogenic RAS and the p53 tumor suppressor** *PLoS Biol* **6**:2853–2868 <https://doi.org/10.1371/journal.pbio.0060301>
- Dimri G. P., Lee X., Basile G., Acosta M., Scott G., Roskelley C., et al. 1995) **A biomarker that identifies senescent human cells in culture and in aging skin in vivo** *Proc Natl Acad Sci U S A* **92**:9363–9367 <https://doi.org/10.1073/pnas.92.20.9363>
- Egan B. M., Pohl F., Anderson X., Williams S. C., Gregory Adodo I., Hunt P., ..., Kornfeld K. 2024) **The ACE inhibitor captopril inhibits ACN-1 to control dauer formation and aging** *Development* **151** <https://doi.org/10.1242/dev.202146>
- Fei M. Y., Wang Y., Chang B. H., Xue K., Dong F. Y., Huang D., ..., Wang L. 2023) **The non-cell-autonomous function of ID1 promotes AML progression via ANGPTL7 from the microenvironment** *Blood* **142**:903–917 <https://doi.org/10.1182/blood.2022019537>
- Genander M., Cook P. J., Ramsköld D., Keyes B. E., Mertz A. F., Sandberg R., Fuchs E 2014) **BMP Signaling and Its pSMAD1/5 Target Genes Differentially Regulate Hair Follicle Stem Cell Lineages** *Cell Stem Cell* **15**:619–633 <https://doi.org/10.1016/j.stem.2014.09.009>
- Gu Z 2022) **Complex heatmap visualization** *Imeta* **1** <https://doi.org/10.1002/imt.2.43>
- Guglielmi L., Heliot C., Kumar S., Alexandrov Y., Gori I., Papaleonidopoulou F., ..., Hill C. S. 2021) **Smad4 controls signaling robustness and morphogenesis by differentially contributing to the Nodal and BMP pathways** *Nature Communications* **12** <https://doi.org/10.1038/s41467-021-26486-3>

- Harrison D. E., Strong R., Allison D. B., Ames B. N., Astle C. M., Atamna H., ..., Miller R. A. 2014) **Acarbose, 17- $\alpha$ -estradiol, and nordihydroguaiaretic acid extend mouse lifespan preferentially in males** *Aging Cell* **13**:273–282 <https://doi.org/10.1111/ace.12170>
- Harrison D. E., Strong R., Sharp Z. D., Nelson J. F., Astle C. M., Flurkey K., ..., Miller R. A. 2009) **Rapamycin fed late in life extends lifespan in genetically heterogeneous mice** *Nature* **460**:392–U108 <https://doi.org/10.1038/nature08221>
- Hayashi Y., Hsiao E. C., Sami S., Lancero M., Schlieve C. R., Nguyen T., ..., Yamanaka S. 2016) **BMP-SMAD-ID promotes reprogramming to pluripotency by inhibiting p16/INK4A-dependent senescence** *Proc Natl Acad Sci U S A* **113**:13057–13062 <https://doi.org/10.1073/pnas.1603668113>
- Janes M. R., Fruman D. A 2009) **Immune Regulation by Rapamycin: Moving Beyond T Cells** *Science Signaling* **2** <https://doi.org/10.1126/scisignal.267pe25>
- Jazwinski S. M 1996) **Longevity, genes, and aging** *Science* **273**:54–59 <https://doi.org/10.1126/science.273.5271.54>
- Johnson A. A., Stolzing A 2019) **The role of lipid metabolism in aging, lifespan regulation, and age-related disease** *Aging Cell* **18** <https://doi.org/10.1111/ace.13048>
- Johnson S. C., Rabinovitch P. S., Kaeberlein M 2013) **mTOR is a key modulator of ageing and age-related disease** *Nature* **493**:338–345 <https://doi.org/10.1038/nature11861>
- Khosla S., Farr J. N., Tchkonja T., Kirkland J. L 2020) **The role of cellular senescence in ageing and endocrine disease** *Nat Rev Endocrinol* **16**:263–275 <https://doi.org/10.1038/s41574-020-0335-y>
- Kim D., Paggi J. M., Park C., Bennett C., Salzberg S. L 2019) **Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype** *Nat Biotechnol* **37**:907–915 <https://doi.org/10.1038/s41587-019-0201-4>
- Kirkland J. L 2016) **Translating the Science of Aging into Therapeutic Interventions** *Cold Spring Harb Perspect Med* **6** <https://doi.org/10.1101/cshperspect.a025908>
- Kumar S., Dietrich N., Kornfeld K 2016) **Angiotensin Converting Enzyme (ACE) Inhibitor Extends Caenorhabditis elegans Life Span** *PLoS Genet* **12** <https://doi.org/10.1371/journal.pgen.1005866>
- Langmead B., Salzberg S. L 2012) **Fast gapped-read alignment with Bowtie 2** *Nat Methods* **9**:357–359 <https://doi.org/10.1038/nmeth.1923>
- Li H., Handsaker B., Wysoker A., Fennell T., Ruan J., Homer N., ..., Genome Project Data Processing, S. 2009) **The Sequence Alignment/Map format and SAMtools** *Bioinformatics* **25**:2078–2079 <https://doi.org/10.1093/bioinformatics/btp352>
- Liang Y. Y., Brunicardi F. C., Lin X 2009) **Smad3 mediates immediate early induction of Id1 by TGF-beta** *Cell Res* **19**:140–148 <https://doi.org/10.1038/cr.2008.321>
- Liao Y., Smyth G. K., Shi W 2014) **featureCounts: an efficient general purpose program for assigning sequence reads to genomic features** *Bioinformatics* **30**:923–930 <https://doi.org/10.1093/bioinformatics/btt656>

- Lihua Z., Claude G., Nathan L 2010) **ChIPpeakAnno: a Bioconductor package to annotate ChIP-seq and ChIP-chip data** *BMC Bioinformatics* **11**
- Lopez-Otin C., Blasco M. A., Partridge L., Serrano M., Kroemer G 2023) **Hallmarks of aging: An expanding universe** *Cell* **186**:243–278 <https://doi.org/10.1016/j.cell.2022.11.001>
- Love M. I., Huber W., Anders S 2014) **Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2** *Genome Biol* **15** <https://doi.org/10.1186/s13059-014-0550-8>
- Lyden D., Young A. Z., Zagzag D., Yan W., Gerald W., O'Reilly R., ..., Benezra R. 1999) **Id1 and Id3 are required for neurogenesis, angiogenesis and vascularization of tumour xenografts** *Nature* **401**:670–677 <https://doi.org/10.1038/44334>
- Lyu G. L., Guan Y. T., Zhang C., Zong L., Sun L., Huang X. K., ..., Tao W. 2018) **TGF- $\beta$  signaling alters H4K20me3 status via miR-29 and contributes to cellular senescence and cardiac aging** *Nature Communications* **9** <https://doi.org/10.1038/s41467-018-06710-3>
- Miyazono K., Maeda S., Imamura T 2005) **BMP receptor signaling: Transcriptional targets, regulation of signals, and signaling cross-talk** *Cytokine & Growth Factor Reviews* **16**:251–263 <https://doi.org/10.1016/j.cytogfr.2005.01.009>
- Munoz-Espin D., Serrano M 2014) **Cellular senescence: from physiology to pathology** *Nat Rev Mol Cell Biol* **15**:482–496 <https://doi.org/10.1038/nrm3823>
- Nadon N. L., Strong R., Miller R. A., Nelson J., Javors M., Sharp Z. D., ..., Harrison D. E. 2008) **Design of aging intervention studies: the NIA interventions testing program** *Age* **30**:187–199 <https://doi.org/10.1007/s11357-008-9048-1>
- Orlowski J 2017) **SMAD5 signaling: more than meets the nuclei** *Cell Res* **27**:1075–1076 <https://doi.org/10.1038/cr.2017.101>
- Radulescu C. I., Cerar V., Haslehurst P., Kopanitsa M., Barnes S. J 2021) **The aging mouse brain: cognition, connectivity and calcium** *Cell Calcium* **94** <https://doi.org/10.1016/j.ceca.2021.102358>
- Rahman M. S., Akhtar N., Jamil H. M., Banik R. S., Asaduzzaman S. M 2015) **TGF- $\beta$ /BMP signaling and other molecular events: regulation of osteoblastogenesis and bone formation** *Bone Research* **3** <https://doi.org/10.1038/boneres.2015.5>
- Ramachandran A., Vizán P., Das D., Chakravarty P., Vogt J., Rogers K. W., ..., Hill C. S. 2018) **TGF- $\beta$  uses a novel mode of receptor activation to phosphorylate SMAD1/5 and induce epithelial-to-mesenchymal transition** *Elife* **7** <https://doi.org/10.7554/eLife.31756>
- Ramirez F., Dundar F., Diehl S., Gruning B. A., Manke T 2014) **deepTools: a flexible platform for exploring deep-sequencing data** *Nucleic Acids Res* **42**:W187–191 <https://doi.org/10.1093/nar/gku365>
- Saunders R. N., Metcalfe M. S., Nicholson M. L 2001) **Rapamycin in transplantation: A review of the evidence** *Kidney International* **59**:3–16 <https://doi.org/10.1046/j.1523-1755.2001.00460.x>
- Schmieder R. E., Hilgers K. F., Schlaich M. P., Schmidt B. M. W 2007) **Renin-angiotensin system and cardiovascular risk** *Lancet* **369**:1208–1219 [https://doi.org/10.1016/S0140-6736\(07\)60242-6](https://doi.org/10.1016/S0140-6736(07)60242-6)

- Senft A. D., Bikoff E. K., Robertson E. J., Costello I 2019) **Genetic dissection of Nodal and Bmp signalling requirements during primordial germ cell development in mouse** *Nature Communications* **10** <https://doi.org/10.1038/s41467-019-09052-w>
- Shah P. P., Donahue G., Otte G. L., Capell B. C., Nelson D. M., Cao K., ..., Berger S. L. 2013) **Lamin B1 depletion in senescent cells triggers large-scale changes in gene expression and the chromatin landscape** *Genes Dev* **27**:1787–1799 <https://doi.org/10.1101/gad.223834.113>
- Strong R., Miller R. A., Antebi A., Astle C. M., Bogue M., Denzel M. S., ..., Harrison D. E. 2016) **Longer lifespan in male mice treated with a weakly estrogenic agonist, an antioxidant, an -glucosidase inhibitor or a Nrf2-inducer** *Aging Cell* **15**:872–884 <https://doi.org/10.1111/accel.12496>
- Strong R., Miller R. A., Cheng C. J., Nelson J. F., Gelfond J., Allani S. K., ..., Harrison D. E. 2022) **Lifespan benefits for the combination of rapamycin plus acarbose and for captopril in genetically heterogeneous mice** *Aging Cell* **21** <https://doi.org/10.1111/accel.13724>
- Sun Y., Yan K., Wang Y., Xu C., Wang D., Zhou W., ..., Xi Q. R. 2022) **Context-dependent tumor-suppressive BMP signaling in diffuse intrinsic pontine glioma regulates stemness through epigenetic regulation of CXXC5** *Nature Cancer* **3**:1105–1122 <https://doi.org/10.1038/s43018-022-00408-8>
- Urist M. R 1965) **Bone - Formation by Autoinduction** *Science* **150**:893–899 <https://doi.org/10.1126/science.150.3698.893>
- Wu T., Hu E., Xu S., Chen M., Guo P., Dai Z., ..., Yu G. 2021) **clusterProfiler 4.0: A universal enrichment tool for interpreting omics data** *Innovation (Camb)* **2** <https://doi.org/10.1016/j.xinn.2021.100141>
- Xu M., Pirtskhalava T., Farr J. N., Weigand B. M., Palmer A. K., Weivoda M. M., ..., Kirkland J. L. 2018) **Senolytics improve physical function and increase lifespan in old age** *Nat Med* **24**:1246–1256 <https://doi.org/10.1038/s41591-018-0092-9>
- Yamamoto K., Rakugi H 2021) **Angiotensin receptor-neprilysin inhibitors: Comprehensive review and implications in hypertension treatment** *Hypertension Research* **44**:1239–1250 <https://doi.org/10.1038/s41440-021-00706-1>
- Ying Q. L., Nichols J., Chambers I., Smith A 2003) **BMP induction of Id proteins suppresses differentiation and sustains embryonic stem cell self-renewal in collaboration with STAT3** *Cell* **115**:281–292 [https://doi.org/10.1016/S0092-8674\(03\)00847-X](https://doi.org/10.1016/S0092-8674(03)00847-X)
- Yousefzadeh M. J., Zhu Y., McGowan S. J., Angelini L., Fuhrmann-Stroissnigg H., Xu M., ..., Niedernhofer L. J. 2018) **Fisetin is a senotherapeutic that extends health and lifespan** *Ebiomedicine* **36**:18–28 <https://doi.org/10.1016/j.ebiom.2018.09.015>
- Zhang Y., Liu T., Meyer C. A., Eeckhoutte J., Johnson D. S., Bernstein B. E., ..., Liu X. S. 2008) **Model-based analysis of ChIP-Seq (MACS)** *Genome Biol* **9** <https://doi.org/10.1186/gb-2008-9-9-r137>
- Zhou Y., Zhou B., Pache L., Chang M., Khodabakhshi A. H., Tanaseichuk O., ..., Chanda S. K. 2019) **Metascape provides a biologist-oriented resource for the analysis of systems-level datasets** *Nat Commun* **10** <https://doi.org/10.1038/s41467-019-09234-6>



Zhu Y., Tchkonja T., Fuhrmann-Stroissnigg H., Dai H. M. M., Ling Y. Y. Y., Stout M. B., ..., Kirkland J. L. 2016) **Identification of a novel senolytic agent, navitoclax, targeting the Bcl-2 family of anti-apoptotic factors** *Aging Cell* **15**:428–435 <https://doi.org/10.1111/acer.12445>

Zhu Y., Tchkonja T., Pirtskhalava T., Gower A. C., Ding H. S., Giorgadze N., ..., Kirkland J. L. 2015) **The Achilles' heel of senescent cells: from transcriptome to senolytic drugs** *Aging Cell* **14**:644–658 <https://doi.org/10.1111/acer.12344>

## Author information

### Wencong Lyu<sup>†</sup>

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China  
ORCID iD: [0009-0001-0456-7560](https://orcid.org/0009-0001-0456-7560)

<sup>†</sup>These authors contributed equally to this work

### Haochen Wang<sup>†</sup>

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

<sup>†</sup>These authors contributed equally to this work

### Zhehao Du

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

### Ran Wei

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

### Jianuo He

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

### Fanju Meng

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

### Jinlong Bi

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

### Lijun Zhang

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

**Chao Zhang**

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

**Yiting Guan\***

Zhanjiang Institute of Clinical Medicine, Central People's Hospital of Zhanjiang, Guangdong Medical University, Zhanjiang, China

ORCID iD: [0000-0001-5158-0226](https://orcid.org/0000-0001-5158-0226)

\*These authors jointly supervised this work

**Wei Tao\***

The MOE Key Laboratory of Cell Proliferation and Differentiation, School of Life Sciences, Peking University, Beijing, China

ORCID iD: [0000-0002-1860-895X](https://orcid.org/0000-0002-1860-895X)

**For correspondence:** [weitao@pku.edu.cn](mailto:weitao@pku.edu.cn)

\*These authors jointly supervised this work

**Editors**

Reviewing Editor

**Weiwei Dang**

Baylor College of Medicine, Houston, United States of America

Senior Editor

**Pankaj Kapahi**

Buck Institute for Research on Aging, Novato, United States of America

**Reviewer #1 (Public review):**

Summary:

In this study, the authors showed that enalapril was able to reduce cellular senescence and improve health status in aged mice. The authors further showed that phosphorylated Smad1/5/9 was significantly elevated and blocking this pathway attenuated the protection of cells from senescence. When middle-aged mice were treated with enalapril, the physiological performance in several tissues, including memory capacity, renal function, and muscle strength, exhibited significant improvement.

Strengths:

The strength of the study lies in the identification of the pSMAD1/5/9 pathway as the underlying mechanism mediating the anti-senescence effects of enalapril with comprehensive evaluation both in vitro and in vivo.

Weaknesses:

The major weakness of the study is the in vivo data. Despite the evidence shown in the in vitro study, there is no data to show that blocking the pSmad1/5/9 pathway is able to attenuate the anti-aging effects of enalapril in the mice. In addition, the aging phenotypes mitigation by enalapril is not evidenced by the extension of lifespan. If it is necessary to show that NAC is able to attenuate enalapril effects in the aging mice. In addition, it would be

beneficial to test if enalapril is able to achieve similar rescue in a premature aging mouse model.

<https://doi.org/10.7554/eLife.104774.1.sa2>

**Reviewer #2 (Public review):**

This manuscript presents an interesting study of enalapril for its potential impact on senescence through the activation of Smad1/5/9 signaling with a focus on antioxidative gene expression. Repurposing enalapril in this context provides a fresh perspective on its effects beyond blood pressure regulation. The authors make a strong case for the importance of Smad1/5/9 in this process, and the inclusion of both in vitro and in vivo models adds value to the findings. Below, I have a few comments and suggestions which may help improve the manuscript.

A major finding in the study is that phosphorylated Smad1/5/9 mediates the effects of enalapril. However, the manuscript focused on the Smad pathway relatively abruptly, and the rationale behind targeting this specific pathway is not fully explained. What makes Smad1/5/9 particularly relevant to the context of this study?

Furthermore, their finding that activation of Smad1/5/9 leads to a reduction of senescence appears somewhat contradictory to the established literature on Smad1/5/9 in senescence. For instance, studies have shown that BMP4-induced senescence involves the activation of Smad1/5/8 (Smad1/5/9), leading to the upregulation of senescence markers like p16 and p21 (JBC, 2009, 284, 12153). Similarly, phosphorylated Smad1/5/8 has been shown to promote and maintain senescence in Ras-activated cells (PLOS Genetics, 2011, 7, e1002359). Could the authors provide more detailed mechanistic insights into why enalapril seems to reverse the typical pro-senescent role of Smad1/5/9 in their study?

While the authors showed that enalapril increases pSmad1/5/9 phosphorylation, what are the expression levels of other key and related factors like Smad4, pSmad2, pSmad3, BMP2, and BMP4 in both senescent and non-senescent cells? These data will help clarify the broader signaling effects.

They used BMP receptor inhibitor LDN193189 to pharmacologically inhibit BMP signaling, but it would be more convincing to also include genetic validation (e.g., knockdown or knockout of BMP2 or BMP4). This will help confirm that the observed effects are truly due to BMP-Smad signaling and not off-target effects of the pharmacological inhibitor LDN.

I don't see the results on the changes in senescence markers p16 and p21 in the mouse models treated with enalapril. Similarly, the effects of enalapril treatment on some key SASP factors, such as TNF- $\alpha$ , MCP-1, IL-1 $\beta$ , and IL-1 $\alpha$ , are missing, particularly in serum and tissues. These are important data to evaluate the effect of enalapril on senescence.

Given that enalapril is primarily known as an antihypertensive, it would be helpful to include data on how it affects blood pressure in the aged mouse models, such as systolic and diastolic blood pressure. This will clarify whether the observed effects are independent of or influenced by changes in blood pressure.

<https://doi.org/10.7554/eLife.104774.1.sa1>

**Author response:**

## Public Reviews:

### Reviewer #1 (Public review):

#### Summary:

*In this study, the authors showed that enalapril was able to reduce cellular senescence and improve health status in aged mice. The authors further showed that phosphorylated Smad1/5/9 was significantly elevated and blocking this pathway attenuated the protection of cells from senescence. When middle-aged mice were treated with enalapril, the physiological performance in several tissues, including memory capacity, renal function, and muscle strength, exhibited significant improvement.*

#### Strengths:

*The strength of the study lies in the identification of the pSMAD1/5/9 pathway as the underlying mechanism mediating the anti-senescence effects of enalapril with comprehensive evaluation both in vitro and in vivo.*

Thanks very much for your insightful evaluation and the constructive suggestions. We have thoroughly studied the comments and a provisional point-to-point response is shown as follows.

#### Weaknesses:

*The major weakness of the study is the in vivo data. Despite the evidence shown in the in vitro study, there is no data to show that blocking the pSmad1/5/9 pathway is able to attenuate the anti-aging effects of enalapril in the mice. In addition, the aging phenotypes mitigation by enalapril is not evidenced by the extension of lifespan.*

Thanks for your comment. As suggested, we will feed LDN193189 to mice while using LDN193189 to block pSmad1/5/9, and will assess age-related phenotypes in the mice to demonstrate that the anti-aging effect of enalapril in mice is mediated through pSmad1/5/9.

We only assess the improvement in the health status of the aging mice, which indicate that enalapril can extend the healthy lifespan of aging mice. This is because we believe that lifespan is controlled by genetics. Therefore, this study focuses solely on the improvement of health phenotypes in aging mice by enalapril.

*If it is necessary to show that NAC is able to attenuate enalapril effects in the aging mice. In addition, it would be beneficial to test if enalapril is able to achieve similar rescue in a premature aging mouse model.*

Thanks for your suggestion. To our knowledge, NAC is an inhibitor of ROS, which is consistent with the antioxidant effect of enalapril. Therefore, we believe that NAC will not diminish the effect of enalapril.

For the premature aging mouse models, we examined the effect of enalapril on Lmna<sup>G609G</sup> mice and other premature aging models and found that the effect was relatively modest. This may be due to differences in the genetic background of premature aging mice, leading to a less pronounced effect of enalapril compared to its impact on naturally aged mice.

### Reviewer #2 (Public review):

*This manuscript presents an interesting study of enalapril for its potential impact on senescence through the activation of Smad1/5/9 signaling with a focus on antioxidative*

*gene expression. Repurposing enalapril in this context provides a fresh perspective on its effects beyond blood pressure regulation. The authors make a strong case for the importance of Smad1/5/9 in this process, and the inclusion of both in vitro and in vivo models adds value to the findings. Below, I have a few comments and suggestions which may help improve the manuscript.*

Thanks very much for your insightful evaluation and the constructive suggestions. We have thoroughly studied the comments and a provisional point-to-point response is shown as follows.

*A major finding in the study is that phosphorylated Smad1/5/9 mediates the effects of enalapril. However, the manuscript focused on the Smad pathway relatively abruptly, and the rationale behind targeting this specific pathway is not fully explained. What makes Smad1/5/9 particularly relevant to the context of this study?*

Thanks for your comment. As stated in the manuscript, after we found that enalapril could improve the cellular senescence phenotype, we screened and examined key targets in important aging-related signaling pathways, such as AKT, mTOR, ERK (Fig. S2A), Smad2/3 and Smad1/5/9 (Fig. 2A). We found that only the phosphorylation levels of Smad1/5/9 significantly increased after enalapril treatment. Therefore, the subsequent focus of this study is on pSmad1/5/9.

*Furthermore, their finding that activation of Smad1/5/9 leads to a reduction of senescence appears somewhat contradictory to the established literature on Smad1/5/9 in senescence. For instance, studies have shown that BMP4-induced senescence involves the activation of Smad1/5/8 (Smad1/5/9), leading to the upregulation of senescence markers like p16 and p21 (JBC, 2009, 284, 12153). Similarly, phosphorylated Smad1/5/8 has been shown to promote and maintain senescence in Ras-activated cells (PLOS Genetics, 2011, 7, e1002359). Could the authors provide more detailed mechanistic insights into why enalapril seems to reverse the typical pro-senescent role of Smad1/5/9 in their study?*

Thanks for your comment. The downstream regulatory network of BMP-pSmad1/5/9 is highly complex. The BMP-SMAD-ID axis has been mentioned in many studies, and its downstream signaling inhibits the expression of p16 and p21 (PNAS, 2016, 113(46), 13057-13062; Cell, 2003, 115(3), 281-292). Additionally, studies have also found that the Smad1-Stat1-P21 axis inhibits osteoblast senescence (Cell Death Discovery, 2022, 8:254). In our study, enalapril was found to increase the expression of ID1, which is a classic downstream target of pSmad1/5/9 (Cell Stem Cell, 2014, 15(5), 619-633). Therefore, pSmad1/5/9 inhibits cellular senescence markers such as p16, p21 and SASP through ID1, thereby promoting cell proliferation (Fig. 3). Furthermore, we also found that pSmad1/5/9 increases the expression of antioxidant genes and reduces ROS levels, exerting antioxidant effects (Fig. 4). Together, ID1 and antioxidant genes enable pSmad1/5/9 to exert its anti-aging effects.

*While the authors showed that enalapril increases pSmad1/5/9 phosphorylation, what are the expression levels of other key and related factors like Smad4, pSmad2, pSmad3, BMP2, and BMP4 in both senescent and non-senescent cells? These data will help clarify the broader signaling effects.*

Thanks for your suggestion. We observed an increase in Smad4 expression, while the levels of pSmad2 and pSmad3 remained unchanged after enalapril treatment (Fig. 2A). We will supplement data on the expression changes of these key factors in both senescent and non-senescent cells.



*They used BMP receptor inhibitor LDN193189 to pharmacologically inhibit BMP signaling, but it would be more convincing to also include genetic validation (e.g., knockdown or knockout of BMP2 or BMP4). This will help confirm that the observed effects are truly due to BMP-Smad signaling and not off-target effects of the pharmacological inhibitor LDN.*

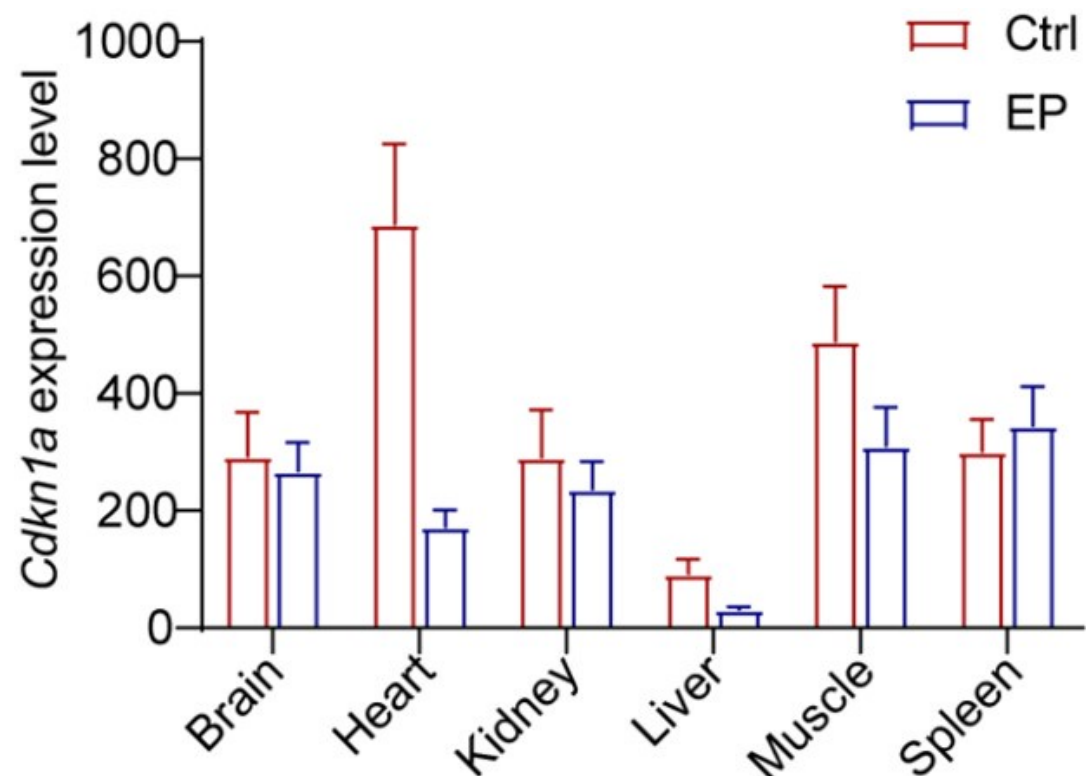
Thanks for your suggestion. We will use shRNA or siRNA to knockdown BMP and examine the related changes to clarify the role of BMP-Smad signaling.

*I don't see the results on the changes in senescence markers p16 and p21 in the mouse models treated with enalapril. Similarly, the effects of enalapril treatment on some key SASP factors, such as TNF- $\alpha$ , MCP-1, IL-1 $\beta$ , and IL-1 $\alpha$ , are missing, particularly in serum and tissues. These are important data to evaluate the effect of enalapril on senescence.*

Thanks for your comment. As for the markers p16 and p21, we observed no change in p16, while the changes in p21 varied across different organs and tissues. (Author response image 1). Nevertheless, behavioral experiments and physiological and biochemical indicators at the individual level consistently demonstrated the significant anti-aging effects of enalapril (Fig. 6).

#### Author response image 1.

p21(*Cdkn1a*) expression levels in organs of mice after enalapril feeding.



We also examined the changes in SASP factors in the serum of mice after enalapril treatment. Notably, SASP factors such as CCL (MCP), CXCL and TNFRS11B showed significant decreases (Fig. 5C). The expression changes of SASP factors varied across different organs. In the liver, kidneys and spleen, the expression of IL1 $\alpha$  and IL1 $\beta$  decreased, while TNFRS11B expression

decreased in both the liver and muscles (Fig. 5B). Additionally, CCL (MCP) levels decreased in all organs (Fig. 5B).

*Given that enalapril is primarily known as an antihypertensive, it would be helpful to include data on how it affects blood pressure in the aged mouse models, such as systolic and diastolic blood pressure. This will clarify whether the observed effects are independent of or influenced by changes in blood pressure.*

Thanks for your comment. We measured the blood pressure in mice, and found no significant change in blood pressure after enalapril treatment, which has also been validated in other studies (J Gerontol A Biol Sci Med Sci, 2019, 74(8), 1149–1157). Therefore, our results are independent of changes in blood pressure.

<https://doi.org/10.7554/eLife.104774.1.sa0>